

Article

A Combined Deep Learning for the Early Detection of Respiratory Disease Using Multiple Environmental

Jake Cowton ^{1,2,*}, Ilias Kyriazakis ² , Thomas Plöt¹ Interdisciplinary Computing and Complex BioSystems, Newcastle University, Newcastle upon Tyne NE1 7RU² Agriculture, School of Natural and Environmental Sciences, Newcastle upon Tyne NE1 7RU, UK; ilias.kyriazakis@newcastle.ac.uk³ Open Lab, Newcastle University, Newcastle upon Tyne

* Correspondence: j.cowton2@newcastle.ac.uk

† Current address: School of Interactive Computing, Georgia

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Abstract: We designed and evaluated an assumption-based animal health monitoring, specifically for the early detection of respiratory disease based on environmental sensor data. Two recurrent neural networks (RNNs), were used to create an autoencoder (AE) model. Environmental data, collected from a variety of sensors, was processed by a type of network trained to reconstruct the patterns of environmental data that did not lead to an occurrence of respiratory disease. A pattern of “healthy environmental data” had a greater likelihood of being reconstructed. Data were labelled as either normal or anomalous using particle swarm optimisation (PSO), from which the model outperformed state-of-the-art techniques, and the actual observations. The results show that a change in the health of pigs showing symptoms of respiratory disease can be detected during time during which their keepers can act to mitigate porcine reproductive and respiratory syndrome (PRRS) in pigs.

Keywords: anomaly detection; deep learning; sensors; environmental data

1. Introduction

The forecasting of an oncoming health challenge is a major level of health and animal welfare. The earlier a disease is detected, thus lowering the overall impact of the disease on the likelihood of treatment success [1]. Respiratory disease syndrome (PRRS), pneumonia, and pleurisy, are so common in commercial pig populations [2]. Therefore, it is important to detect respiratory disease, which we define as the point at which the disease increases from 0 to greater than 0.

There are many factors that influence the conditions of the environment which the pigs inhabit. A number of factors, such as [2]—specifically, air quality, temperature,

文章一种深度学习GRU自动编 统疾病的早期检测，采用多

Jake Cowton ^{1,2,*}, Ilias Kyriazakis ² , Thomas Plöt¹ 跨学科计算与复杂生物系统研究组，计算机学院，纽卡斯尔大学，纽卡斯尔，英国；jaume.bacardit@newcastle.ac.uk² 农业学院，自然与农业学院，纽卡斯尔大学，纽卡斯尔，英国；ilias.kyriazakis@newcastle.ac.uk³ 开放大学，thomas.ploetz@gatech.edu* 通讯作者：j.cowton

† 当前地址：交互计算学院，佐治亚理工学院，亚特兰大

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摘要：我们设计并评估了一种无假设的、基于深度传感器数据的生长猪呼吸系统疾病的早期检测。该方案使用了一种神经网络（RNN），用于构建一个自动编码器（AE）模型，以检测异常。自动编码器是一种经过训练以重建疾病发生的环境数据训练的GRU-AE，不符合“健康”模式的所有重建误差均使用基于阈值的异常检测方法（该模型正常或异常，从而触发警报。GRU-AE方法的结果在出现偏差时，会触发警报。结果表明，环境变化可在猪出现呼吸系统疾病（如猪繁殖与呼吸综合征（PRRS）（一种病毒性疾病）的负面影响之前，其饲养者有一段时间

关键词：异常检测；深度学习；传感器；门控循环神经网络

1. 引言

预测动物即将面临的健康挑战是维持高水平健康养殖，从而降低疾病对动物和农场整体的影响，并提高养殖效率。猪繁殖与呼吸综合征（PRRS）、肺炎和胸膜炎，是商业养殖中常见的疾病。检测呼吸系统疾病的发生具有极高的价值，我们将其定义为疾病发生的点。

影响呼吸系统疾病发生发展的因素有很多，特别是环境因素中，[2]——具体包括空气质量、温度和湿度——的连续监测。

g GRU-Autoencoder for piratory Disease in Pigs ntal Sensors

Plötz ^{3,†}  and Jaume Bacardit ¹ 

stems (ICOS) Research Group, School of Computing,
7RU, UK; jaume.bacardit@newcastle.ac.uk
al Sciences, Newcastle University,
kis@newcastle.ac.uk
n Tyne NE1 7RU, UK; thomas.ploetz@gatech.edu

, Georgia Institute of Technology, Atlanta, GA 30318, USA.

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mption-free, deep learning-based methodology for
rly detection of respiratory disease in growing pigs
ent neural networks (RNNs), each comprising gated
autoencoder (GRU-AE) into which environmental
rocessed to detect anomalies. An autoencoder is a
rns it is fed as input. By training the GRU-AE using
rence of respiratory disease, data that did not fit the
reater reconstruction error. All reconstruction errors
sing threshold-based anomaly detection optimised
hich alerts are raised. The results from the GRU-AE
s, raising alerts when such predictions deviated from
change in the environment can result in occurrences
se within 1–7 days, meaning that there is a period of
ate the negative effect of respiratory diseases, such as
(PRRS), a common and destructive disease endemic

ensors; GRU; pig; health; disease

nge in animals is fundamental to maintaining a high
disease is predicted, the sooner it can be dealt with,
n both the animal and the farm, and increasing the
iseases, such as porcine reproductive and respiratory
some of the most common types of disease found
is highly valuable to be able to predict occurrences
oint in time where respiratory disease prevalence

contraction of respiratory diseases, in particular,
er of the respiratory disease risk factors discussed by
ture, and humidity—are now able to be monitored

码器组合用于猪呼吸系 种环境传感器

lötz ^{3,†}  and Jaume Bacardit ¹ 

卡斯尔大学, 纽卡斯尔 NE17RU, 英国;
与环境科学学院, 纽卡斯尔大学, 纽卡斯尔 NE1
放实验室, 纽卡斯尔大学, 纽卡斯尔 NE17RU, 英
owton2@newcastle.ac.uk

兰大, GA 30318, 美国

: 2018年8月2日



度学习的动物健康监测方法, 特别针对基于环境传
方法使用了两个包含门控循环单元 (GRU) 的循环
GRU-AE), 将来自各种传感器的环境数据输入其
建其输入模式的网络。通过使用不会导致呼吸系统
康环境数据”模式的数据会产生更大的重建误差。
该检测方法经过粒子群优化算法PSO优化) 标记为
果优于最先进技术, 当预测结果与实际观察结果出
在1-7天内导致猪出现呼吸系统疾病症状, 这意味着
(PRRS), 这是一种在猪中常见且具有破坏性的地
间可以采取行动减轻这种疾病的影响。

环单元; 猪; 健康; 疾病

康和动物福利的基础。越早预测疾病, 就能越早处
高治疗成功的可能性 [1]。呼吸系统疾病, 如猪繁
业猪群中最常见的疾病类型之一 [2]。因此, 能够预
定义为呼吸系统疾病患病率从0增加到大于0的时间

是猪所生活的环境。Stark 等人讨论的呼吸系统疾病
——现在能够通过廉价的电子传感器进行实时、高分

continuously in real time and at a high resolution used to further understand the relationship between

In most situations, recording data continuously normal circumstances, as anomalous events are less generated allow us to train a solid model of normality quantify any deviations from it.

The objective of this paper is to describe and of anomalies pertaining to respiratory disease in A recurrent autoencoder, built using deep learning reconstruct raw sensor data which does not lead to pigs. This GRU-autoencoder (GRU-AE) can then be used to how close it is to normality. A threshold-based anomaly swarm optimisation (PSO) used this score to determine too great and therefore represent environmental conditions trained to represent.

The benefits of using such a method are that typically required in classical statistical approaches minimal preprocessing. Also, no arbitrary threshold is wholly data-driven. In addition, there is no relationship between environmental conditions and network. This work is structured as an anomaly detection increases in respiratory disease prevalence; therefore of low disease prevalence without affecting performance are, and the higher the prevalence at these times, the

Malhotra et al. [3] demonstrated the potential encoder-decoder structure for multi-sensor anomaly potential for using data which is not easily predicted rather than a standard artificial neural network (ANN) dimension, and used reconstruction error was used their implementation on various datasets suffered (Section 4.1.1). This paper revisits the design choices et al., specifically by:

- exchanging long short-term memory cells for
- introducing particle swarm optimisation to optimise the loss from the GRU-AE is anomalous
- changing how the overall system is evaluated detection system

The remainder of the paper is structured as approach taken. Section 3 explains the composition of followed by Section 4, which details how each of the performance of the systems described, followed by drawn in Section 7.

2. Background

In recent years, agricultural research has begun statistical approaches and started to regularly incorporate such as Support Vector Machines (SVM) and Random analysis [5–10].

Recent work has seen SVMs being used, for curve of eggs produced by commercial hens, covering

这些数据可用于进一步理解环境条件与呼吸系统疾病

在大多数情况下，连续记录数据意味着存储大量低。生成的正常数据量很大，使我们能够训练一个坚固然后量化任何偏离正常性的情况。

本文的目的是描述和评估一种基于环境因素，对循环自动编码器（使用深度学习（门控循环单元（GRU）吸系统疾病发生的一批猪的原始传感器数据。然后，常性的接近程度对传感器数据窗口进行“评分”。一常检测器使用此分数来确定是否对过高分数（因此代报。

使用这种方法的优点在于，无需手工设计特征，传感器数据可以在最少预处理的情况下使用。此外，数据驱动的。另外，无需深入了解环境条件与猪健康这项工作被构建为一个异常检测解决方案，而不是例能够在由低患病率数据组成的集合上运行，而不会影率越高，就越容易微调 and 验证该方法。

Malhotra等人 [3] 展示了基于长短期记忆单元检测中的潜力，表明该方法对于使用难以预测的数据元，而不是标准的人工神经网络（ANN）结构，以而，他们在各种数据集上的实现结果在精确率或召回新审视了先前工作中的设计选择，特别是Malhotra

- 用门控循环单元（GRU）细胞替换长短期记忆（LSTM）
- 将粒子群优化算法引入以优化异常检测器，该检测
- 改变整体系统的评估方式，以展示更均衡的异常检

本文的其余部分结构如下。首先，第2节介绍了及所应用的预处理，随后是第4节，该节详细介绍了性能，随后是第6节的讨论以及第7节得出的结论。

2. 背景

近年来，农业研究开始逐渐摆脱单纯依赖经典统计支持向量机（SVM）和随机森林分类器，纳入数据

近期研究已看到支持向量机被用于检测商业母鸡以实现早期预警系统的目的

sing inexpensive electronic sensors. Such data can be en environmental conditions and respiratory disease. sly means storing data that predominantly describes less frequent. The large amounts of normal data ality. Hence, it is better to model normality and then

nd evaluate a methodology for the early warning in growing pigs, based on environmental factors. ning (gated recurrent units (GRU)), learns how to o occurrences of respiratory disease within a batch of e used to “score” a window of sensor data according anomaly detector that was optimised using particle rmine whether to raise an alert for scores which are conditions that do not fit in line with the data it was

at there is no need for hand-crafted features, as is es, meaning that raw sensor data can be used with holds are set, as the detection of anomalous events requirement for an in-depth understanding of the nd pig health, as this is handled implicitly within the etection solution, rather than, for example, predicting ore, the method is able to perform on data consisting ormance. However, the more cases of disease there the easier it is to fine-tune and validate the method. ntial of a long short-term memory (LSTM)-based aly detection, showing the methodology has good ictable [4]. This implementation used LSTM cells, (ANN) structure, in order to handle the temporal ed to determine anomalies. However, the results of ed distinctly either in precision or recall (defined in es made in previous work, such as that by Malhotra

r GRU cells

ptimise an anomaly detector that determines whether

uated to demonstrate a more balanced anomaly

as follows. Firstly, Section 2 gives context for the n of the data and the preprocessing that was applied, the models were constructed. Section 5 outlines the by the discussion in Section 6 and the conclusions

egun to shift away from relying only on classical ncorporate machine learning (ML) methodologies, ndom Forest Classifiers, into the approach to data

or example, to detect anomalies in the production ering a range of prediction windows for the purpose

病之间的关系。

量主要描述正常情况的数据，因为异常事件频率较坚实的正常模型。因此，最好先对正常性进行建模，

对生长猪呼吸系统疾病进行早期预警的方法。一种 GRU) 构建))，学习如何重建不会导致猪群中呼，该GRU自动编码器 (GRU-AE) 可以根据其与正一个使用粒子群优化 (PSO) 优化的基于阈值的异代表不符合其训练数据所代表的环境条件) 发出警

，正如经典统计方法通常所要求的，这意味着原始，不会设置任意阈值，因为异常事件的检测完全是康之间的关系，因为这一点在网络内部被隐式处理。例如预测呼吸系统疾病患病率增加；因此，该方法影响性能。然而，病例越多，以及这些时间的患病

元 (LSTM) 的编码器-解码器结构在多传感器异常据具有良好的潜力 [4]。该实现使用了长短期记忆单便处理时间维度，并使用重建误差来确定异常。然回率 (定义在Section 4.1.1) 上明显受损。本文重ra等人所做的工作，具体包括：

STM) 细胞

测器用于判断GRU-AE的损失是否异常

检测系统

了所采用方法的背景。第3节解释了数据的组成以了每个模型的构建方法。第5节概述了所描述系统的

统计方法，转而定期将机器学习 (ML) 方法，如分析方法中 [5–10]。

鸡产蛋量生产曲线中的异常，覆盖了多种预测窗口，

of an early warning system [5]. Other research has models) for the prediction of avian influenza prevalence performance over support vector regression [6].

Classification methods have been used to see implementation of one-class classifiers (OCCs). SVM-though there are many potential methods which have high sensitivity but a comparatively poor specificity imbalance in the datasets that these methodologies are not always suitable, particularly if the true negatives. Cabral et al. compared a number of methods for detection for one-class classification (FBDOCC) of heart

There are also approaches to anomaly detection and correlation analysis [12]. Clustering uses distance from a cluster centroid exceeds an acceptable value. However, methods falling into this category are typically of low quality of analysis and speed due to the inherent complexity of analysis and speed due to the inherent complexity of analysis uses similarity metrics to determine if new

Anomaly detection can be used in health applications to detect deviations from normality. A common methodology has been shown to outperform classical statistical models, and ANOVA regression [14]. In one simulation network was able to correctly trigger an alarm within an average model and 11–12 days for the others. Anomaly often there are not necessarily certain conditions that certain conditions. For example, when detecting anomalies it is not possible to have knowledge of all of the different conditions. It is therefore necessary to understand what normality

In health applications where sensors are used there is a need to focus not only on a high detection rate, but also on high, alerts from this system will not be trusted. Sensors to be detected in near real time [18] in order to provide a reliable service. This creates a substantial limitation resource intensive.

Often, these approaches require a transformation of useful feature representation [19–21]. However, deep learning this step. Raw data can be passed to a deep neural network is handled intrinsically by the network [16]. How can a function effectively [22]. This is particularly valuable as a neural network is deep enough to model the data, all data

A relatively small amount of work in agriculture has used deep learning, which mostly focuses on image processing demonstrates how deep learning methods, used in machine learning used for time-series analysis for the purpose of anomaly

的早期预警系统 [5]。其他研究则应用集成学习方法显示出比支持向量回归更好的性能 [6]。

分类方法已通过单类分类器 (OCCs) 的实施将 SVM 方法在此方法中非常常见, 尽管有许多潜在方法在特异性相对较差 [8], 考虑到这些方法通常用于的数据指标并不总是适用, 特别是当真正负例的数量远远超过的 OCCs, 并提出了用于心脏疾病单类分类 (FBDO

还有分类之外的异常检测方法, 如聚类 [11] 和相类质心的距离是否超过可接受值: 如果是这样, 则被析质量和速度之间取得合理平衡, 这是由于时间序列来确定新数据是否与已知正常数据相似。

异常检测可用于健康应用, 以检测疾病爆发或其是贝叶斯网络, 研究表明其性能优于经典统计方法, 在一项炭疽疫情模拟中 [15], 贝叶斯网络能够在 2 他方法则需要 11–12 天。异常检测非常适合医疗保健是偏离某些条件。例如, 通过心电图 (ECG) 检测心可能会出现新类型 [16]。因此, 有必要了解正常状态

在传感器用于健康应用的情况下, 存在两个主要关注低误报率 [17], 因为如果误报率高, 系统发出的如此, 但异常检测应在近乎实时 [18] 进行, 以便快来了重大限制, 因为某些方法非常资源密集。

通常, 这些方法需要对数据进行转换, 以便生成的方法则无需这一步骤。原始数据可以直接输入深度内在功能的一部分 [16]。但这也需要大量数据才能价值, 只要网络足够深能够对数据进行建模, 所有数

农业领域相对较少的工作利用了新兴技术, 如深度学习本文展示了深度学习方法与进化算法结合如何用于时

as applied ensemble learning methods (combined ensemble using meteorological data, showing improved

separate normality from abnormality through the M-based methods are very common in this approach, which can be applied. Models of this type usually have specificity [8], which is to be expected given the natural imbalances are commonly used for. However, these metrics negatively drastically outweigh the number of false positives on OCCs with the proposal of features boundary for heart disease [8].

tation outside of classification, such as clustering [11] distance metrics to determine if a new point's distance value: if this is the case, it is considered anomalous. It is typically unable to strike a reasonable balance between characteristics of time-series data [13]. Correlation with data is similar to known normal data.

applications to detect outbreaks of disease or other anomaly. One of the primary methods used for this task are Bayesian networks, which are statistical methods, such as control charts, moving average and Kalman filter. In the case of an anthrax outbreak [15], the Bayesian network was able to detect the outbreak within 2 days, in comparison to 9 days for the moving average method. Anomaly detection is very well suited to healthcare, as it can detect deviations from normality that constitute a problem, but rather deviations from normality through electrocardiography (ECG), it is not clear what types, as new types can occur in the future [16]. Anomaly detection looks like, rather than what problems look like. However, there are two main challenges. Firstly, there is a trade-off between a high detection rate, but also a low false positive rate [17], as, if this is not the case, then the system will be overwhelmed. Secondly, though not in all cases, anomalies ought to be detected as quickly as possible so that interventions can be taken on which methods can be used, as some are very

difficult to be carried out on the data in order to generate a deep learning-based methods benefit by not requiring manual work without any hand-crafting of features, as this is not the case, however, this requires a substantial amount of data to be available for high-dimensional data, as, provided the data can be utilised.

The future has made use of emerging technologies, such as machine learning analysis and remote sensing [23]. This paper in conjunction with evolutionary algorithms, can be used for anomaly detection.

法（组合模型）利用气象数据预测禽流感流行情况，

将正常性与异常性分离。基于支持向量机的（SVM）方法可以应用。此类模型的灵敏度通常很高，但数据集的自然不平衡，这是可以预料的。然而，这些模型在超过假正例的数量时。Cabral等人比较了多种常见OCC）的特征边界检测 [8]。

相关分析 [12]。聚类使用距离度量来确定新点与聚类中心被视为异常。然而，属于此类的方法通常无法在分列数据的固有特性 [13]。相关分析使用相似性度量

其他偏离正常的情况。用于此任务的一种常见方法，例如控制图、移动平均模型和ANOVA回归 [14]。2天内正确触发警报，而移动平均模型需要9天，其健康领域，因为通常并非存在构成问题的特定条件，而心律失常时，不可能了解所有不同类型，因为未来状态是什么样子的，而不是问题状态是什么样子的。

要挑战。首先，不仅需要关注高检测率，还需要关注警报将不被信任。其次，虽然并非所有情况下都能快速响应，从而采取干预措施。这给可使用的方

成有用的特征表示 [19–21]。然而，基于深度神经网络，无需手工设计特征，因为特征提取是网络有效运行 [22]。对于高维数据而言，这一点尤其有数据都能被利用。

深度学习，这些技术主要关注图像分析和遥感 [23]。时间序列分析，目的是进行异常检测。

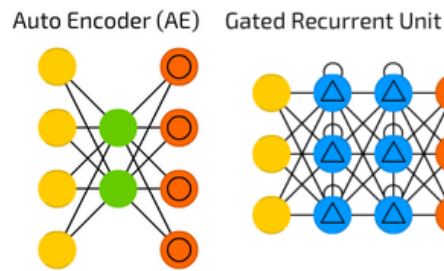


Figure 1. A visualisation of a classic autoencoder (left) and a Gated Recurrent Unit (right) [24]. These two autoencoder (AE) capable of utilising temporality.

Autoencoders are artificial neural networks (ANNs) (Figure 1). Given an input tensor N , an encoding M is trained to minimise the loss between N and N' . the attempted reconstruction of M to N' constitute performance in assisting anomaly detection, often using it as the input to an OCC [25–27], though some to detect anomalous data [28].

3. Experimental Design

3.1. Data Collection

Data was collected for growing pigs from a varied multidisciplinary project. Each farm was represented where a batch consists of varying population sizes for varying lengths of time (14–145 days). The start weight was 17.5 to 27.6 kg. The number of pigs per batch affected the basis. Respiratory disease prevalence was divided into Individual Pig Care [29]. A pig could be either scored or treated for each level of severity. Health data regarding disease was collected by a trained person. They made that showed symptoms of certain diseases and clinical symptoms fell. Given the nature of the system, pigs number of pigs that are classified as showing symptoms as such in the previous day.

The environmental data was collected using Ge the centre of the room in which the batch was located relevant to pig health [31]. Briefly, the system monitored +250 °C $\pm$ 0.05%, relative humidity (a sensor sensitive -40 °C to $+85$ °C), and CO₂ concentration (with a range of reading, operating temperature range from 0 °C was sampled every minute.

3.2. Data Preprocessing

Outliers were determined to be anything outside from the data. Missing data induced both from the were filled using linear interpolation. This was due to short-term variance, meaning that, on average, the data was reasonable. The results of this preprocessing of

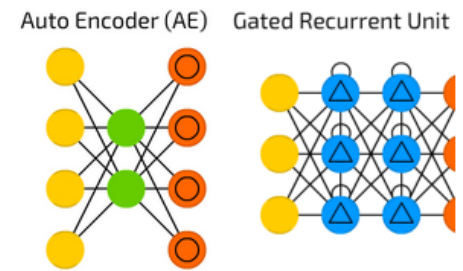


图1。一个经典自编码器（左）和由门控循环单元（GRU）（右）。这两种方法是我们将它们结合在一起以创建一个能够

自编码器是人工神经网络（ANNs）的一种，它输入张量 N ，找到一个编码 M ，用于创建 N' 。该网络为 M 是编码器，而将 M 尝试重建为 N' 则构成解码器，通常通过学习特征表示 M 并将其用作OCC [25–27] 检测异常数据 [28]。

3. 实验设计

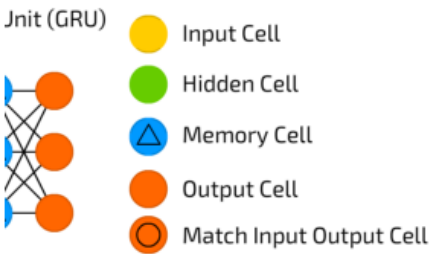
3.1. 数据收集

数据是作为一项欧盟资助的多学科项目的一部分，每个农场都由一批或多批猪（3–16批）组成，一批包含次被监测了不同的时间长度（14–145天）。猪的初始中受呼吸系统疾病影响的猪的数量每天都会被记录。如Zoetis的个体猪护理 [29]所定义。一头猪可以被评分个严重程度级别都如此。有关显示疾病症状的猪数量统计了每批中显示某些疾病症状的猪的数量，并根据的性质，每天无法单独识别猪，因此被归类为显示症状的猪。

环境数据是使用通用警报 [30] 传感器收集的，这记录了与猪健康相关的环境特征 [31]。简而言之，该系统监测了相对湿度（传感器灵敏度为 $\pm 2\%$ ，工作温度范围从 0 ppm至 5000 ppm ± 30 ppm或 $\pm 3\%$ 的读数，每分钟采样一次。

3.2. 数据预处理

离群值被定义为任何超出 $\mu \pm 2\sigma$ 每个变量每个值移除和硬件问题导致的缺失数据均使用线性插值填充，这意味着平均而言，两个值之间的非线性变化所示。三个传感器



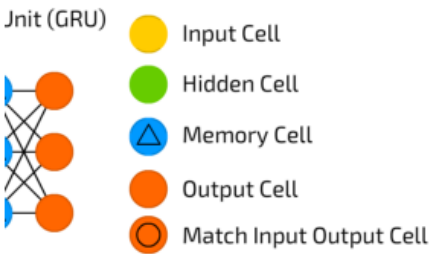
er (left) and recurrent neural network consisting of
o methodologies are what we combined to create an
y.

ANNs) that are trained to reconstruct input patterns
 M is found which is used to create N' . The network
' . The transformation of N to M is the encoder, and
utes the decoder. Autoencoders have shown good
n through learning the feature representation M and
me papers have made use of the reconstruction error

ariety of farms across Europe as part of an EU-funded
nted by a number of batches of pigs (3–16 batches),
es (40–1294 pigs), and the batches were monitored
rting weight of the pigs was variable, ranging from
cted by respiratory disease was recorded on a daily
d into four levels of severity, as defined by Zoetis's
ored (identified as diseased but no treatment given)
egarding the number of pigs showing symptoms of
manually counted the number of pigs in each batch
classified it by the level of severity into which the
gs cannot be identified individually each day, so the
ptoms can be new cases or pigs that were identified

General Alert [30] sensors, which were suspended in
ted. The sensors recorded the environmental features
onitored temperature (a sensor range of $-50\text{ }^{\circ}\text{C}$ to
sitivity of $\pm 2\%$ and operating temperature range of
a range from 0 ppm to 5000 ppm ± 30 ppm or $\pm 3\%$
 $^{\circ}\text{C}$ to $50\text{ }^{\circ}\text{C}$, and humidity from 0% to 95%), which

side $\mu \pm 2\sigma$ per variable per batch and were removed
the removal of outliers and due to hardware issues
s deemed acceptable, as the environment has low
e assumption of a linear change between two values
f sensor data are shown in Figure 2. The three sensor



(GRU) 组成的循环神经网络 (右) 的示意图 [24]。
够利用时序性的自编码器 (AE) 。

它被训练用于重建输入模式 (图 1)。给定一个输
络被训练以最小化 N 和 N' 之间的损失。将 N 转换
码器。自编码器在辅助异常检测方面表现出良好性
25–27], 的输入——尽管一些论文利用重建误差来检

分，从欧洲各地的多种农场收集的用于生长猪。每
含不同的人口规模 (40-1294头猪)，并且这些批
始体重是可变的，范围从17.5到27.6公斤。每批猪
。呼吸系统疾病的发病率被分为四个严重程度级别，
评分 (被确定为患病但未给予治疗) 或被治疗，每
量的健康数据由一名训练有素的人员收集。他们手
根据症状所属的严重程度级别进行分类。鉴于系统
症状的猪数量可以是新病例或前一天被确定为这样

这些传感器悬挂在存放批次的房间中央。传感器记
系统监测温度 (传感器范围为 $-50\text{ }^{\circ}\text{C}$ 至 $+250\text{ }^{\circ}\text{C} \pm 0.1$
温度范围为 $-40\text{ }^{\circ}\text{C}$ 至 $+85\text{ }^{\circ}\text{C}$)，以及 CO_2 浓度 (范
，工作温度范围为 $0\text{ }^{\circ}\text{C}$ 至 $50\text{ }^{\circ}\text{C}$ ，湿度从0%至95%)，

批次范围之外的数据，并从数据中移除。由于离群
填充。这被认为是可接受的，因为环境具有低短期
化假设是合理的。传感器数据的预处理结果如图2

readings—temperature, humidity, and CO2—constituted the four severity levels were summed and used to calculate the prevalence of respiratory disease on a given day, rather than looking at the number of pig

$$\text{prevalence} = \frac{\text{Number of pig}}{\text{Total}}$$

This was done because respiratory diseases rarely occur in most cases they are treated before the disease develops. There were very few cases of severe or irrecoverable respiratory disease. After aggregation of data, the level of prevalence was still low (Figure 3). The sensor data was merged with the data from the latter by duplicating the values for the full 24 h period.

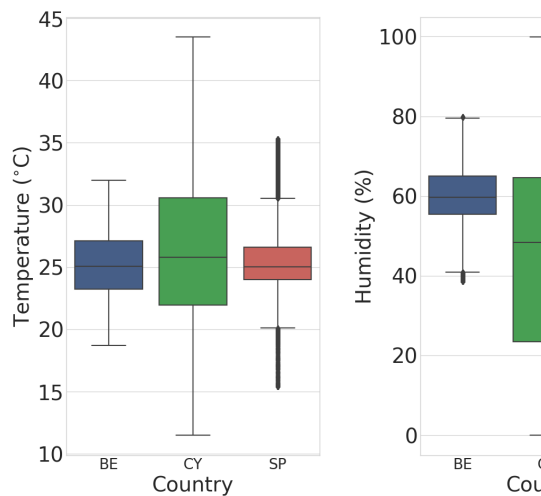


Figure 2. A boxplot describing the temperature, humidity, and CO2 included in the data collection—Belgium (BE), Cyprus (CY), and Spain (SP).

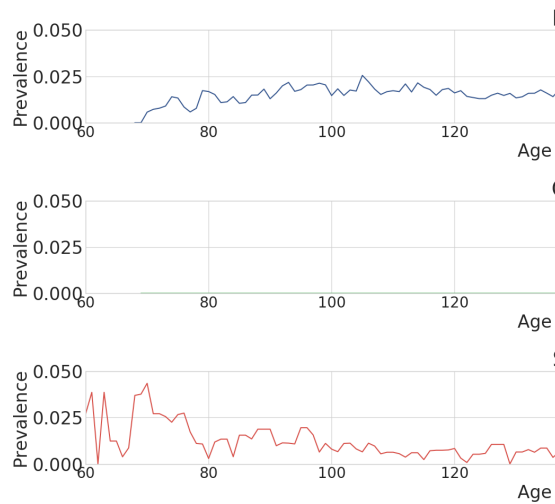


Figure 3. A graph showing the average prevalence of respiratory disease in growing pigs in Belgium (BE), Cyprus (CY), and Spain (SP).

读数——温度、湿度和CO2——构成第4节中概述的模型。每天表现出呼吸系统疾病症状的猪的比例，而不是查看

$$\text{prevalence} = \frac{\text{Number of pig}}{\text{Total}}$$

这是因为在大多数情况下，呼吸系统疾病很少在出现症状之前通常已被治疗。因此，所有农场中严重或不可逆的数据聚合，患病率水平仍然非常低，并且随着时间的增加（每日健康数据）的分辨率上采样，通过复制值来到1之间。

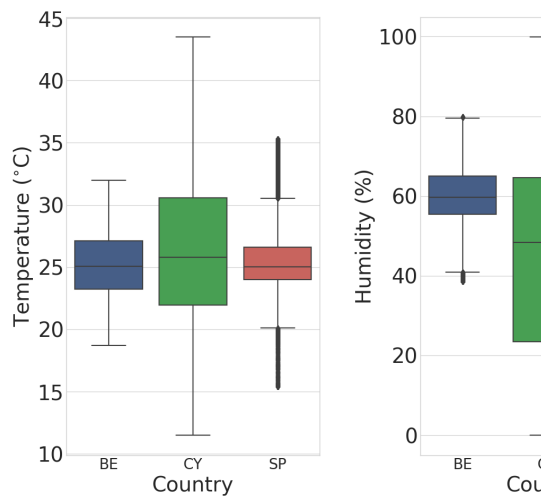


图2。 一个箱线图，描述了温度、湿度和CO2 传感器（BE）、塞浦路斯（CY）和西班牙（SP）——在预

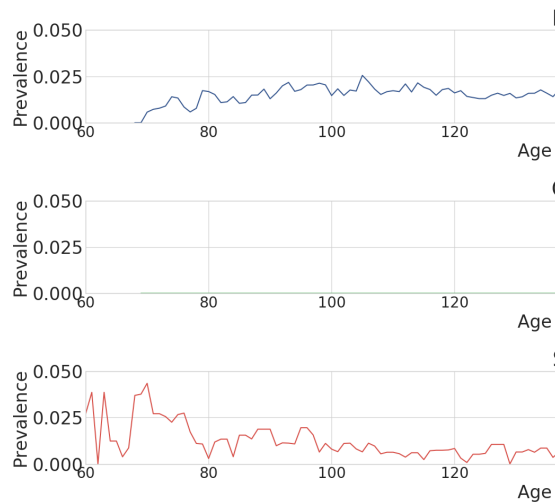


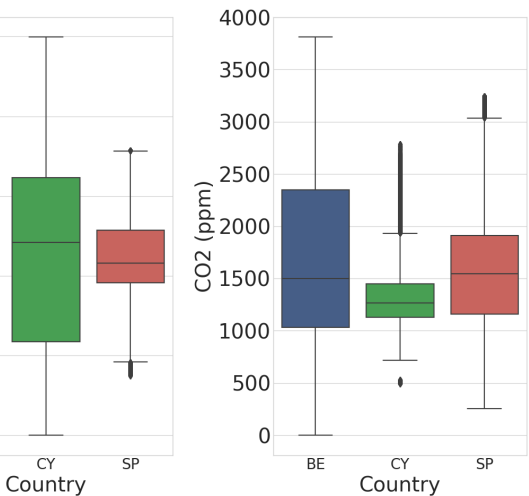
图3。 一个图表，显示了呼吸系统疾病在比利时（BE）批次的平均患病率与年龄（以天为单位）的关系，在

stitute the inputs to the models outlined in Section 4. calculate the proportion of pigs showing symptoms looking at one specific level of severity (Equation (1)).

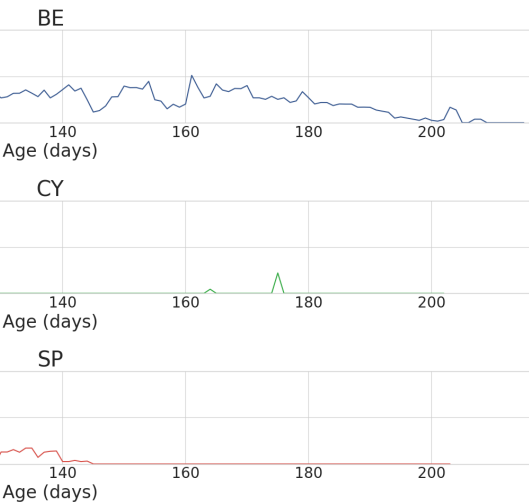
pigs showing any symptoms
tal number of pigs

(1)

rarely exceed the classification of light or mild, as e develops into more severe symptoms. Therefore, le respiratory disease across all farms. Even with the till very low, and it decreased even further over time daily health data by upscaling the resolution of the eriod. All input was normalised to between 0 and 1.



umidity, and CO₂ sensor data for the three countries yprus (CY), and Spain (SP)—after preprocessing.



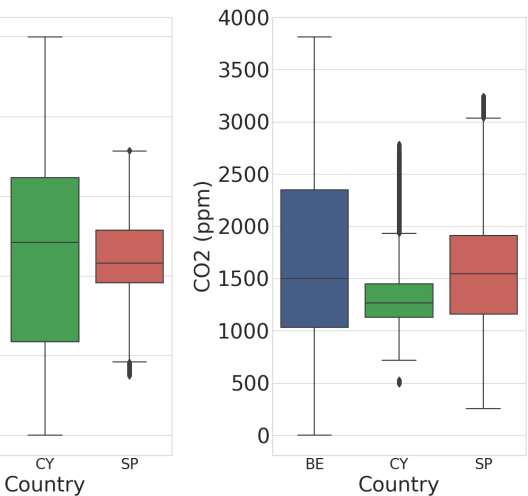
ce of respiratory disease against age (in days) in all (CY), and Spain (SP) after preprocessing.

模型的输入。四种严重程度被汇总并用于计算某一 看特定的严重程度级别（公式（1））。

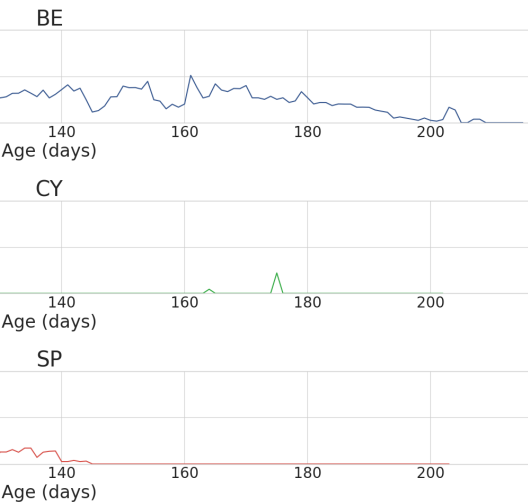
pigs showing any symptoms
tal number of pigs

(1)

超过轻症或轻度的分类，因为疾病在发展成更严重 或不可逆转的呼吸系统疾病病例非常少。即使通过 的推移进一步降低（图3）。传感器数据通过将后 来合并全天的24小时数据。所有输入均归一化到0



器数据，针对数据收集中包含的三个国家——比利时 预处理后的情况。



BE）、塞浦路斯（CY）和西班牙（SP）所有生长猪 在预处理后。

3.2.1. Separation of Assumed Healthy and Assume

As this paper investigates the effects of enviro the existence of a lag period between a change in en environment will not immediately result in an outbr were both evaluated [32]; greater windows were no of data available for training. This lag period was u . Using this window, the data was labelled as on unhealthy". Data from any point where the proporti to u days after the disease prevalence decreases to 0 was labelled "assumed healthy". All data at this poi sliding window.

3.2.2. Splitting Data into Train, Validation, and Test

It was necessary to split data into training and t was used to achieve this (Figure 4). Our method, (GRU-R) model used for comparison, detailed in S GRU network and a threshold-based anomaly detec required two sets of data for the full workflow. H_t , H were used for the anomaly detector. H_t was used fo to validate it. O_e was used to optimise an anomaly detector was tested on to attain an evaluation metric

O_t was a single, independent farm batch, O_e w H_v contained only the assumed healthy data of an only the assumed healthy data from all the remaini

Each of the four datasets could be represented minutes of data in a farm batch, and each X_n cont CO_2 . This matrix was broken down into frames usi timepoint (1 min) per step, creating $T \in \mathbb{R}^{m \times w \times 3}$, w

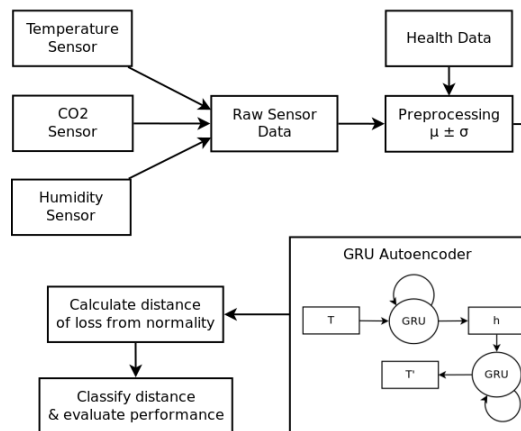


Figure 4. An overview of the entire processing workflow for anomaly detection. Temperature, CO_2 , and humidity sensors data. This was then broken down into four folds—(used for the anomaly detection). Each of these folds at 30 min. Each window of 30 min was processed produced. The distance between this reconstruction the window is anomalous or not, using a threshold-swarm optimisation.

3.2.1. 假设健康数据与假设异常数据的分离

由于本文研究了环境对健康的影响，因此有必要环境变化不会立即导致疾病爆发 [32]。评估了14天和会大大减少可用于训练的数据量。这个滞后期被称为两个集合之一：“假设健康”和“假设不健康”。在的猪的比例大于0的任何时间点被标记为“假设不健时所有数据都使用滑动窗口分解为30分钟的“帧”。

3.2.2. 将数据分割为训练集、验证集和测试集

需要将数据分割为训练集和测试集。采用留一批第4.1节，以及用于对比的GRU回归（GRU-R）模型基于阈值的异常检测器。GRU网络和异常检测器在GRU网络，而 O_e 和 O_t 用于异常检测器。 H_t 用于GRU子群优化算法优化异常检测器，而 O_t 是用于获得评何重叠。

O_t 是一个单一、独立的农场批次， O_e 由三个随次的假定健康数据，而 H_t 仅包含所有剩余农场批次

每个数据集都可以表示为一个矩阵 $X \in \mathbb{R}^{n \times 3}$ ，含温度、湿度和 CO_2 的传感器数据。该矩阵使用大点（1分钟），创建 $T \in \mathbb{R}^{m \times w \times 3}$ ，其中 m 是集合

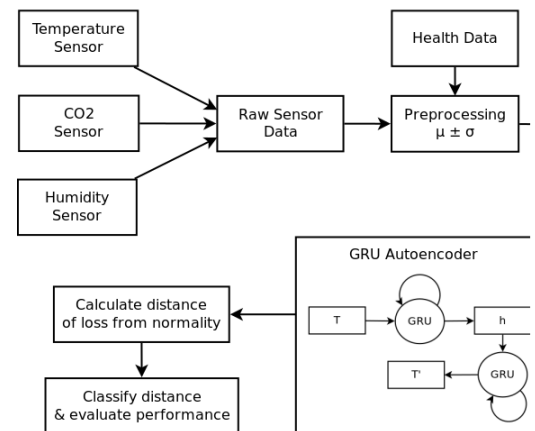


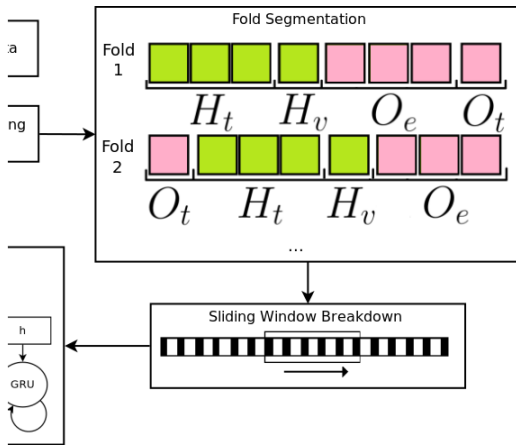
图 4. 整个处理工作流程的概述，从原始传感器数据并合并到每日健康数据中。然后将其分解为四个折叠检测)。这些折叠中的每一个都使用30分钟的滑动窗并产生一个重建误差。这个重建误差与正常值的距离异常检测器进行检测，该检测器使用粒子群优化进行

ed Unhealthy Data

environment on health, it was necessary to account for environment and a change in health, as a change in tbreak of disease [32]. Lag periods of 14 and 21 days not used, as this would greatly reduce the amount as referred to as the assumed unhealthy window, one of two sets: “assumed healthy” and “assumed ortion of pigs showing symptoms was greater than 0 0 was labelled “assumed unhealthy”; all other data oint was broken down into 30 min “frames” using a

st Sets

d testing sets. Leave-one-batch-out cross-validation d, detailed in Section 4.1, and the GRU-regression Section 4.3.3, were each composed of two parts: a tector. Both the GRU network and anomaly detector H_t, H_v were used for the GRU network, and O_e and O_t for the training of GRU networks and H_v was used ly detector via PSO and O_t was the set the anomaly tric. There was no overlap between any of these sets. e was composed of three randomly chosen batches, another independent farm batch, and H_t contained ining farm batches. ted as a matrix $X \in \mathbb{R}^{n \times 3}$, where n is the number of ntains sensor data for temperature, humidity, and using a sliding window of size w , which moved one , where m was the number of frames in the set.



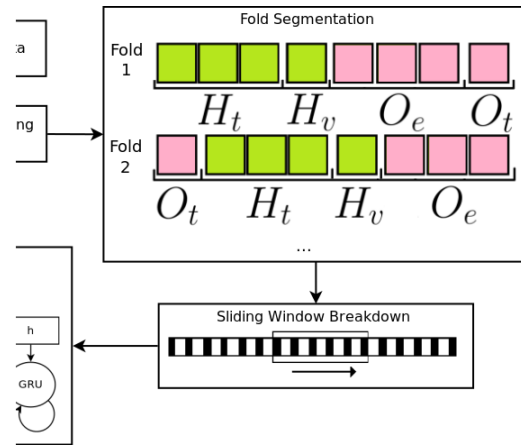
g workflow, from the raw sensor data to anomaly ors were preprocessed and merged with daily health — H_t and H_v (used for the GRU-AE), and O_e and O_t lds were broken down using a sliding window sized ed by the GRU-AE, and a reconstruction error was ion error and normality was used to detect whether ld-based anomaly detector optimised using particle

要考虑环境变化与健康变化之间存在滞后期，因为和21天的滞后期 [32]；未使用更长的窗口，因为这为假设不健康窗口 u 。使用此窗口，数据被标记为在疾病患病率下降到0之后的0到 u 天内，出现症状健康”；所有其他数据被标记为“假设健康”。此。

批次交叉验证方法实现（图4）。我们的方法，详见型，详见第4.3.3节，均由两部分组成：GRU网络和完整工作流程中均需要两组数据。 H_t 、 H_v 用于RU网络的训练，而 H_v 用于验证它。 O_e 用于通过粒估指标的异常检测器测试集。这些集合之间没有任

随机选择的批次组成， H_v 仅包含另一个独立农场批次的假定健康数据。

³，其中 n 是农场批次中数据分钟数，而每个 X_n 包大小为 w 的滑动窗口分解为帧，每次移动一个时间合中的帧数。



据到异常检测。温度、CO₂和湿度传感器被预处理叠— H_t 和 H_v (用于GRU-AE)，和 O_e 和 O_t (用于异常窗口进行分解。每个30分钟的窗口由GRU-AE处理，离被用于检测窗口是否异常，使用一个基于阈值的行优化。

3.2.3. Preliminary Experiments Utilising Batch Normalisation

Batch normalisation is a deep learning technique for making neural networks more robust by performing a per layer normalisation. The meaning of batch in this context differs from the meaning in a standard manuscript. In this context, batches are the subsets of the network being trained with the subsequent back-detection, the effect of this technique is not necessarily based on preliminary experiments, was found to be that batch normalisation allows a model to deal with inter-batch variance by its mean and variance [33], meaning that the model has not seen before. The decrease in performance when using batch normalisation is a requirement that the model be highly sensitive to shifts in the final models.

4. Time-Series Early Warning Methods

4.1. GRU-Autoencoder

The implementation of an autoencoder model consists of two networks—an encoder and a decoder—that were trained on traditional ANN structure (Figure 1), which does not require the encoder to learn a fixed-size representation h of an input. The encoder learned a fixed-size representation h of an input x measured by using it to attempt a reconstruction of x to minimise the reconstruction error, calculated using T was reversed when calculating loss, as it creates shared information between performance in sequence-to-sequence networks [34].

H_t was used to train the GRU-AE using mini-batches contained only “assumed healthy” data, which resulted in a lower reconstruction error for this class of data. Hyper-parameters, hidden size, number of layers, and momentum were selected using grid search, a reconstruction error for each fold's H_v were selected. Information the GRU-AE can store for each time series is a large amount of memory for this task, the hidden size is expected to be fairly small. The number of recurrent layers creates deep models and understanding of the time series and memory. The mini-batch size is how many sequences before back-propagating the loss. This is typically described though is entirely dependent on the other parameters. It is a regularisation technique used to reduce the likelihood of overfitting in artificial neural networks. The learning rate and momentum are hyper-parameters. They determine how drastically weights are updated. Hyper-parameters were then used to calculate reconstruction errors.

Once all reconstruction errors were calculated, to determine whether each reconstruction error of a given frame e and the distribution of the reconstruction errors R , where $\Sigma = \text{COV}(e, R)$. This was then smoothed using a Gaussian kernel. If the smoothed distance was above some threshold (Section 4.1.1), then it was considered an anomalous frame.

3.2.3. 批量归一化初步实验

批量归一化是一种深度学习技术，通过在输入数据网络方面非常有效。请注意，这里的批量与本文其余批次是训练数据的小块，这些小块被分批输入到正在训练的模型中。在异常检测中，这种技术的影响不一定是理想的损害性能。这是因为批量归一化允许模型通过对其均方误差意味着模型更有能力处理它以前从未见过的数据。敏感于数据中的微小变化，因此它没有被用于最终模型。使用批量归一化时性能下降表明，模型没有被用于最终模型。

4. 时间序列早期预警方法

4.1. GRU自动编码器

自动编码器模型（图4）的实现由两个多层门控循环单元（LSTM）被同时训练。这不同于传统的ANN结构（图1）输入 h 的固定大小表示 T 。该表示的质量通过使用最小化重建误差，该误差使用二元交叉熵损失函数因为它创建了短期依赖关系，这导致序列到序列网络。

H_t 被用于使用小批量RMSprop梯度下降 [35]训练。这导致GRU-AE针对此类数据优化出较低的重建误差。超参数、隐藏层大小、层数、小批量大小、Dropout等参数在每个折的交叉验证上产生最低的重建误差。给定的时间序列可以存储的信息量。由于这项任务大小预期会相对较小。层数表示GRU创建深度模型很昂贵。小批量大小是指在反向传播损失之前通过网络。Dropout完全取决于其他参数和可用的内存量。Dropout并且在标准人工神经网络中很常见。学习率和动量也影响传播期间权重的变化幅度。然后使用选定的超参数计算误差。

一旦所有重建误差都计算完毕，阈值异常检测器通过计算给定帧的重建误差（2）与所有帧的重建误差使用局部加权平滑（LOWESS）[36]对距离进行平滑化方法在4.1.1节中描述），则该帧被视为异常帧。

ormalisation

ique that has shown to be very effective at training malization process on input data. Please note that the farm batches used throughout the rest of the ets of the training data that are fed in blocks to the k-propagation-based error correction. In anomaly ssarily desirable and, in this particular application, o be detrimental to performance. This was because ternal covariate shift by normalising each mini-batch model is more capable of handling data it has never using batch normalisation showed that there is a slight variations in the data, and so it was not used

odel (Figure 4) consisted of two multi-layer GRU ere trained simultaneously. This is instead of the not intrinsically account for temporality in data. The an input T . The quality of this representation was of T using the decoder. The model was optimised using binary cross-entropy loss, between T and T' . s short-term dependencies that result in an increased [34].

i-batch RMSprop gradient descent [35]. This dataset h resulted in the GRU-AE optimising towards a ta and a higher reconstruction error for all other of layers, mini-batch size, dropout, learning rate, , and the parameters which resulted in the lowest lected. The hidden size represents the amount of series it is given. As there was no requirement for a size representation identified by the grid search was rrent layers represents the capability of the GRU to series. These are expensive in terms of computation sequences are passed forward through the network desired to be as large as possible to speed up training, eters and the amount of memory available. Dropout ikelihood of over-fitting and is common in standard omentum are also used in standard artificial neural s are changed during back-propagation. The selected onstruction errors for all frames in O_e and O_t .

ed, the threshold-based anomaly detector was used of a frame was to be considered anomalous or not. is distance (Equation (2)) between the reconstruction reconstruction errors for all frames of H_v , defined as d using locally weighted smoothing (LOWESS) [36]. shold τ (the optimisation of which is described in us frame.

数据上执行逐层归一化过程, 已被证明在训练鲁棒余部分使用的农场批次含义不同。在这种情况下, 在训练的网络中, 随后通过基于反向传播的错误校的, 并且在本次特定应用中, 根据初步实验发现会均值和方差进行归一化来处理内部协变量偏移, 这。使用批量归一化时性能下降表明, 模型需要高度模型。[33], 这意味着模型更有能力处理它以前从未模型需要高度敏感于数据中的微小变化, 因此它没

控循环单元网络组成——一个编码器和一个解码器, 1), 后者在数据中不内在地考虑时序性。编码器学用它来尝试使用解码器重建 T 来衡量。该模型被优计算得出, 用于 T 和 T' 。 T 在计算损失时被反转, 络性能提高 [34]。

]训练GRU-AE。该数据集仅包含“假设健康”数据, 差, 而对于所有其他数据则产生较高的重建误差。 out、学习率和动量是使用网格搜索选择的, 并且选低重建误差的参数。隐藏层大小表示GRU-AE为每务对内存没有大量需求, 网格搜索识别出的隐藏层和时间序列理解的能力。这些在计算和内存方面都网络的序列数量。这通常希望尽可能大以加快训练, out是一种正则化技术, 用于降低过拟合的可能性, 也在标准人工神经网络中使用。它们决定了在反向计算 O_e 和<样式id='11'>O</样式> $_t$ 中所有帧的重建

器被用于判断每帧的重建误差是否应被视为异常。 误差分布 e 对于 H_v , 即 R , 其中 $\Sigma = \text{COV}(e, R)$ 。然后滑处理。如果平滑后的距离高于某个阈值 τ (其优

$$d = \sqrt{(e - \bar{R})^T}$$

$$d = \sqrt{(e - \bar{R})^T}$$

4.1.1. PSO-Optimised Anomaly Detection

Once all distances were calculated for all frames, anomaly detector, capable of determining which PSO [37]—a metaheuristic, which uses concepts of s that produce a near-optimal solution. Consider the sequence $\bar{D} = \{d_0, d_1, d_2, \dots, d_n\}$, such that each reconstruction error of a frame at index i from R . The validity of an alert was assessed by determining within an acceptable forecasting window starting at an alert had been raised for this increase in prevalence, t positive. Furthermore, consider the vector $\vec{F} = \{f_0, f_1, \dots, f_n\}$ of animals showing symptoms of respiratory disease any frame i : $f_{i-1} = 0 \wedge f_i > 0$ for which there is no

The parameters τ , α , and β of the detector were then tested on O_t for a final performance evaluation with three dimensions representing τ , α , and β . This was designed to maximise both Matthews correlation at the same time, minimising β . α and β were not fitness function.

$$F = 2 \cdot \frac{TP \times T}{(TP + FP)(FP + F)}$$

MCC is a measure designed to evaluate the performance of a classifier in problems with high class imbalance, s classification, -1 for perfect misclassification, and F-measure is commonly used in information/document retrieval, as they are not important to the correctly classifying negatives (true negatives) is an

It was desirable to attain a high α , as this corresponds to the disease is raised, whereas β needs to be minimised, a For example, given the values $\alpha = 7$ and $\beta = 1$, we know that occurrences of respiratory disease will appear within 7–8 days, a As we consider MCC to be the most important component in Equation 3.

4.2. Other Metrics Used for Evaluation

Specificity, a commonly used evaluation metric for a large number of negative examples (the typical specificity, we have used positive predictive value (also known as recall, Equation (6)), which we identify

4.1.1. PSO优化异常检测

一旦所有帧的距离都计算完毕，下一步是使用P哪些帧应被视为异常。PSO [37]是一种元启发式算法产生接近最优的解决方案。考虑序列中所有帧的马氏距离 d_i 是帧索引 i 处重建误差 R 的马氏距离。定义警报为从开始、 $\alpha + \beta$ 结束的可接受预测窗口内是否从0增加因疾病流行率增加而触发，则该警报被视为真阳性， $f_n\}$ ，其中每个 f_i 包含帧 i 显示呼吸系统疾病症状的 $f_i > 0$ ，在 $i - \alpha$ 和 $i - \beta$ 之间没有警报的情况。

检测器的参数 τ , α , 和 β , 通过PSO算法进行对每个参数的最佳值在 O_t 上进行最终性能评估。 α , 和 β . PSO使用的适应度函数(公式(3))旨同时最小化 β . α 和 β 在输入适应度函数时被归一

$$F = 2 \cdot \frac{TP \times T}{(TP + FP)(FP + F)}$$

MCC是一个用于评估二元分类器预测的指标，中特别稳健。MCC在完美分类时得分为1，在完美错F-measure常用于信息/文档检索，因为它不会受到 negatives) 在计算中。在文档检索中这不是问题，统中，正确分类负例(真负例)是系统的核心目标之

希望能够获得较高的 α ，因为这对应于疾病预警持续的时间。例如，给定 $\alpha = 7$ 和 $\beta = 1$ 的值，我作将在7-8天内出现。对于不同的检测器，给定 $\alpha = 3$ 和 $\beta = 5$ ，我们计呼吸系统疾病的发作将在3-8天内出现，这是一个MCC是我们适应度函数最重要的组成部分，它在公

4.2. 其他用于评估的指标

特异性，一个常用的评估指标，在包含大量负例生误导性分数。我们未使用特异性，而是采用了阳性起见，本文将此指标称为精确率。我们还使用了灵敏为召回率。我们没有使用

$$\bar{R})^T \Sigma^{-1} (e - \bar{R}) \quad (2)$$

es, the next step was to optimise the threshold-based h frames should be considered anomalous, using f swarm intelligence to find a set of hyper-parameters the vector of Mahalanobis distances of all frames each d_i is the Mahalanobis distance between the . An alert was defined as any $d_i : d_{i-1} < \tau \wedge d_i \geq \tau$. ining whether disease prevalence increased from 0 g at $i + \alpha$ and ending at $\alpha + \beta$. If so, and if no other e, the alert was considered a true positive, else a false $0, f_1, f_2, \dots, f_n\}$ such that each f_i contains the number ase at frame i . A false negative was determined to be o alert within $i - \alpha$ and $i - \beta$.

r were the parameters to be optimised using PSO. ndently in O_e , and the best values for each parameter evaluation. Each particle was randomly initialised The fitness function (Equation (3)) used by the PSO lation coefficient MCC (Equation (4)) and α whilst, ormalised between 0 and 1 when entered into the

$$CC + \alpha - \beta \quad (3)$$

$$\frac{TN - FP \times FN}{+ FN)(TN + FP)(TN + FN)} \quad (4)$$

predictions of binary classifiers, which is especially , such as our dataset. MCC scores 1 for perfect nd 0 for equivalent to random classification. The cument retrieval, as it does not fall prey to the same atives in the calculation. This is not an issue within that particular task; however, in a detection system, an integral objective of the system.

esponds to how far in advance the alert of oncoming d, as this represents the period of time the alert is for. we know that when an alert is raised, it is expected ear within 7–8 days. Given a different detector for an alert is raised, it is expected that occurrences of , a larger window and, therefore, more uncertainty. ponent of our fitness function, it has a larger weight

tric, can produce misleading scores in datasets with al scenario in anomaly detection tasks). Instead of (also known as precision, Equation (5)). For the sake oughout this paper. We also made use of sensitivity ntify as recall throughout this paper. We did not use

$$\bar{R})^T \Sigma^{-1} (e - \bar{R}) \quad (2)$$

PSO [37]优化阈值异常检测器，该检测器能够判断法，它利用群智能的概念来找到一组超参数，从而氏距离向量 $\vec{D} = \{d_0, d_1, d_2, \dots, d_n\}$ ，其中每个任何 $d_i : d_{i-1} < \tau \wedge d_i \geq \tau$ 。通过确定疾病在 $i + \alpha$ 来评估警报的有效性。如果是，并且没有其他警报，否则为假阳性。此外，考虑向量 $\vec{F} = \{f_0, f_1, f_2, \dots\}$ ，的动物数量。假阴性被定义为任何帧 $i : f_{i-1} = 0 \wedge$

进行优化。每个折独立地使用粒子进行优化 O_e ，然。每个粒子被随机初始化为三个维度，分别代表 τ ，在最大化马修斯相关系数MCC（公式（4））和 α ，一化到0和1之间。

$$CC + \alpha - \beta \quad (3)$$

$$\frac{TN - FP \times FN}{+ FN)(TN + FP)(TN + FN)} \quad (4)$$

，在具有高度类别不平衡的问题（如我们的数据集）错误分类时得分为 -1 ，在随机分类时得分为0。到相同问题的困扰，但它不包括真负例（true，因为它们对该特定任务不重要；然而，在检测系之一。

警提前程度，而 β 则需要最小化，因为这代表预我们知道当预警被触发时，预计呼吸系统疾病的发 $= 3$ 和 $\beta = 5$ 的值，我们知道当预警被触发时，预个更大的窗口，因此不确定性也更大。由于我们认为公式 3. 中具有更大的权重。

例（异常检测任务中的典型场景）的数据集中会产生性预测值（也称为精确率，公式（5））。为简洁敏度（也称为召回率，公式（6）），本文将其称

precision or recall for the optimisation of the hyper-p evaluate the final performance of each method.

$$P = \frac{tp}{tp + fp}$$

$$R = \frac{tp}{tp + fn}$$

4.3. Methods Included for Comparison

All methods used were tested on identical test carried out, a Wilcoxon test was conducted to compare three comparison methods and the GRU-AE. These using Holm's method [38].

4.3.1. Luminol

In order to provide a baseline for comparison of time-series anomaly detection package, Luminol [39] anomaly detection, as well as methods which require component analysis (PCA). This uses a bitmap representing symbolic aggregate approximation (SAX) data [40]—to detect anomalies [41]. This is implemented using a window and a leading window, which slide across the SAX representation and subsequently into bitmap. The distance between the two bitmaps is calculated within the window. The calculated anomaly scores are then tested by the anomaly detector in the same way that the GRU-environmental variables into a univariate time series.

4.3.2. Time-Series Regression with Autoregression I

Autoregression integrated moving average (ARIMA) is a time-series analysis in statistical analysis. The parameters are p , the degree of differencing; and q , the order of the moving average. A grid search on H_t , optimising for the lowest Akaike Information Criterion (AIC) for comparing the goodness of fit of statistical models. Once the optimal parameters were found, H_v was used to predict the next 30-minute windows, given the previous 180 minutes between the predicted window and ground truth. An alert for the window $i + \alpha$ to $i + \alpha + \beta$. Values of α and β were tested on O_t . Similar to the Luminol method, the three environmental variables were transformed into a univariate time series.

4.3.3. Time-Series Regression with GRU Network (GRU-AE)

To provide context for how the GRU-AE performs, a GRU network was used for regression of the environmental sensors. By training the GRU network, it learned to predict environmental variables for a respiratory disease. The model was trained to predict the next 30-minute windows used for the GRU-AE. Similar to the ARIMA model, values beyond some threshold (τ) were considered an

precision or recall to optimize the hyperparameters of the detector. They are only used for

$$P = \frac{tp}{tp + fp}$$

$$R = \frac{tp}{tp + fn}$$

4.3. 用于比较的方法

所有使用的方法都在相同的测试集上进行了测试。我们使用了Wilcoxon检验，以比较三种比较方法与GRU-AE。Holm方法进行了多重观察校正 [38]。

4.3.1. Luminol

为了提供性能比较的基线，LinkedIn的开源、这个库代表了开箱即用的异常检测，以及需要数据为输入。这使用单变量时间序列的位图表示，该表示使用离散化方法 [40]——来检测异常 [41]。这是使用两个窗口，它们在时间序列上滑动。每个窗口然后被转换为位图。计算两个位图之间的距离，并代表领先窗口。GRU-AE相同的方式使用网格搜索优化的基于阈值的单变量时间序列，取了第一个主成分。

4.3.2. 时间序列回归与自回归积分移动平均 (ARIMA)

自回归积分移动平均 (ARIMA) 是统计分析中一种参数模型，参数包括 p (自回归模型的阶数)、 d (差分阶数) 以及 q (移动平均的阶数)。通过寻找使赤池信息量标准 (AIC) 最低的参数组合，找到最优参数后，使用 H_v 对给定前180分钟在 O_e 中的数据。预测窗口与真实值之间的窗口 $i + \alpha$ 到 $i + \alpha + \beta$ 的警报。 τ 、 α 和 β 的值与 Luminol 方法类似，取第一主成分将三个环境变量

4.3.3. 使用GRU网络的时间序列回归 (GRU-AE)

为了提供关于GRU-AE相对于另一种深度学习网络被训练以预测环境传感器未来的值。通过仅使用健康会导致呼吸系统疾病发生的条件下的环境变量。该模型窗口大小相同。类似于ARIMA模型，均方误差超过 $\tau + \alpha + \beta$ 的警报。类似于

r-parameters of the detector. They were only used to

$$\frac{tp}{tp + fp} \quad (5)$$

$$\frac{tp}{tp + fn} \quad (6)$$

test sets O_t . Once all methods for comparison were
mpare the precision, recall, and MCC between the
ese results were corrected for multiple observations

of performance, LinkedIn's open source, univariate
39], was used. This library represents out-of-the-box
ire data be univariate, often achieved using principal
epresentation of a univariate time series, calculated
)—a method for the discretisation of time-series
mented using two concatenated windows, a lagged
s the time series. Each window is then converted into
aps based on the frequency of each SAX subword.
ed and represents the anomaly score for the leading
ested using a grid search-optimised threshold-based
U-AE is assessed. In order to transform the three
ries, the first principal component was taken.

n Integrated Moving Average (ARIMA)

(ARIMA) is a frequently used model for univariate
ameters for this model (p , the order of autoregressive
der of the moving average model) were found using
ike information criteria (AIC). This is a metric used
odels, though it is not a measure of goodness of fit.
used to fit the model. This was then used to forecast
utes in O_e . Windows where the mean square error
h was beyond some threshold (τ) were considered
s for τ , α , and β were optimised using grid search
method, the first principal component was taken to
a univariate time series.

(GRU-R)

rforms with regards to an alternative deep learning
ession. This network was trained to predict future
g this regression network on only healthy data H_t ,
r conditions which do not result in occurrences of
redict windows of 30 minutes, the same size as the
IMA model, windows where the mean square error
an alert for the window $i + \alpha$ to $i + \alpha + \beta$. Similar to

于评估每种方法的最终性能。

$$\frac{tp}{tp + fp} \quad (5)$$

$$\frac{tp}{tp + fn} \quad (6)$$

试 O_{t_0} 。一旦所有用于比较的方法都完成了，就进行
E之间的精确率、召回率和MCC。这些结果使用

、单变量时间序列异常检测包Luminol [39], 被使用。
为单变量的方法，通常通过主成分分析（PCA）实
用符号聚合近似（SAX）计算——一种时间序列数
两个连接的窗口实现的，一个滞后窗口和一个领先窗
为SAX表示，并随后基于每个SAX子词的频率转换
口的异常分数。计算出的异常分数然后使用与评估
的异常检测器进行测试。为了将三个环境变量转换

MA)

中用于单变量时间序列分析的常用模型。该模型的
 q ，移动平均模型的阶数）通过在 H_t 上进行网格搜
这是一个用于比较统计模型拟合优度的指标，但它
对模型进行拟合。然后使用它来预测30分钟窗口，
之间的均方误差超过某个阈值 (τ) 的窗口被视为对
通过网格搜索优化，最后在 O_t 上进行测试。与
量转换为单变量时间序列。

U-R)

方法性能的背景，使用GRU网络进行了回归。该网
健康数据 H_t 训练这个回归网络，它学会了预测不
模型被训练以预测30分钟窗口，与GRU-AE使用的
过某个阈值 (τ) 的窗口被考虑为窗口 $i + \alpha$ 到 i

the configuration laid out in Section 4.1, these param using O_e , and finally evaluated on O_t .

5. Results

For the purpose of exploration, the GRU-AE’s grid search with a fixed $\beta = 6$ with O_v used as the te a steady and sustained decrease in the total number the point where the acceptable forecast window ext increased portion of alerts that were unverifiable in the maximum range of the acceptable forecast windo to reduce the search space of the PSO, and, therefor The search space was also restricted by stipulating following ranges: $0 \leq \tau \leq 20$, $1 \leq \alpha \leq 12$, and $1 \leq$ this was found through preliminary testing. β was li this was deemed to be too large.

As outlined in Section 4.1, grid search was use The grid search converged to the same set of para space, we report in Table 1 the most frequently identi the hidden size was very low. This is because unde in the environment, and, therefore, only a small am deviations from this—frequent or drastic fluctuati be expected.

Table 1. The grid search-optimised hyper-pa multidimensional time-series data.

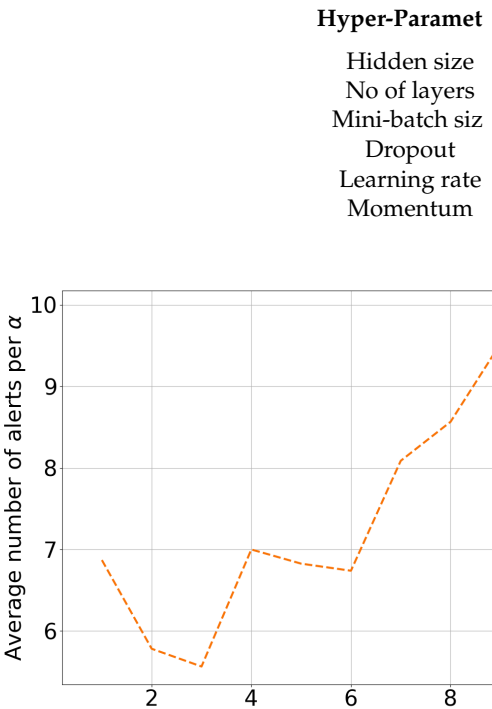


Figure 5. The average number of alerts raised, giv $\beta = 7$, using the optimal threshold τ for each given α used on Mahalanobis distances produced by a GRU

第4.1节中概述的配置，这些参数使用粒子群优化算最终在 O_t 上进行评估。

5. 结果

为探索目的，GRU-AE的异常检测器最初使用网从 $\alpha = 10$ 开始，警报总数呈稳定且持续的下降（图5围，导致部分警报在有效性方面无法验证。警报的减该最大端点用于缩小PSO的搜索空间，因此，如果 α 规定如果粒子超出以下范围，则也被视为无效： $0 \leq$ 涵盖了所有可行值；这是通过初步测试发现的。 β

如第 4.1节所述，采用网格搜索方法确定了GRU到相同的参数集，因此，为节省篇幅，我们在表 1 中隐藏层大小非常小。这是因为健康状态下环境波动很时——频繁或剧烈的波动——就可以预期出现呼吸系

表1。用于训练多维时间序列数据的GRU-AE的网格

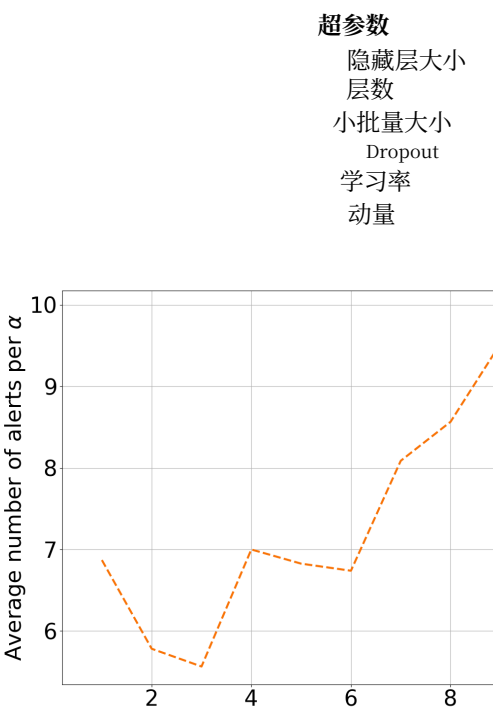


图5。给定警报窗口起始时间为 α 天和大小为 $\beta = 7$ 氏距离异常检测上找到的针对每个给定的 α 的最优阈

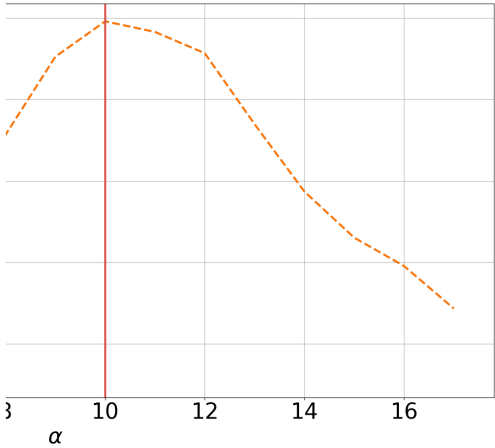
ameters were optimised using PSO, which is trained

's anomaly detector was initially optimised using a test set. For values from $\alpha = 10$ onwards, there was ber of alerts raised (Figure 5). This was indicative of extended beyond the data available, resulting in an in terms of validity. This decline in alerts signified dow $\alpha + \beta = 16$. This maximum endpoint was used fore, a particle was considered invalid if $\alpha + \beta > 16$. ng that a particle was also invalid if it exceeded the $\leq \beta \leq 6$. The limits on τ cover all feasible values; s limited to a maximum of 6, as a range greater than

used to find the hyper-parameters for the GRU-AE. arameters in almost all batches, so, for the sake of ntified values for all hyper-parameters. As expected, der healthy circumstances there is little fluctuation amount of memory is required. It is when there are ations—that occurrences of respiratory disease can

-parameters used to train a GRU-AE used on

eter	Value
ze	4
rs	20
size	1024
	0.5
ate	1×10^{-6}
m	0.3



given an alert window start of α days and a size of n α found using grid search on the anomaly detection RU-based autoencoder.

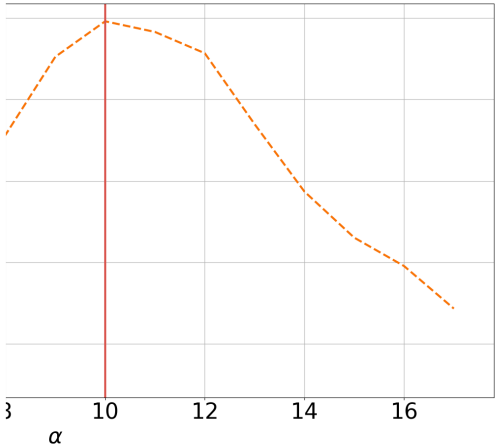
算法PSO进行了优化，该算法使用 O_e 进行训练，并

网格搜索进行了优化，其中固定使用 O_v 作为测试集。图5)。这表明可接受的预测窗口已超出可用数据范减少表明可接受的预测窗口的最大范围 $\alpha + \beta = 16$ 。 $\alpha + \beta > 16$ ，则粒子被视为无效。搜索空间还通过 $\leq \tau \leq 20$ 、 $1 \leq \alpha \leq 12$ 和 $1 \leq \beta \leq 6$ 。对 τ 的限制限制为最大6，因为大于此范围的值被认为太大。

U-AE的超参数。网格搜索在几乎所有批次中都收敛中报告了所有超参数出现频率最高的值。正如预期，很小，因此只需要少量内存。当出现偏离这种情况系统疾病。

格搜索优化超参数。

值
4
20
1024
0.5
1×10^{-6}
0.3



7时，平均触发的警报数量，使用通过网格搜索在马阈值 τ ，该异常检测由基于GRU的自动编码器产生。

Table 2 shows the batch-by-batch performance presents the performance of the three compared s. The GRU-AE achieves better results than the other (precision, recall, and MCC). The most notable difference methodologies is the substantial shift in performance achieved this, but to a much lower degree. In the L achieved in comparison to recall, indicating that th. However, in the GRU-AE model, the solutions found of precision, $\mu_P = 0.776$, $\mu_R = 0.938$, whilst not sacrifice overall performance. The substantially higher recall capacity to identify negative changes in the environment with symptoms of respiratory disease; the lower precision negative changes in the environment result in occurrence the fact that there are other factors [42] that contribute environmental measures used in this analysis.

The last row of Table 2 contains a high-level methods across batches. We have counted the number of an incorrect score of 0 for MCC (I), correct score of 1 of the C count, Luminol clearly performs the worst score (10–11). At the same time, GRU-AE stands out the score of ARIMA and GRU-R, and almost four times keeping the lowest count of B.

Numerically, at least, we can see that the difference may be considered substantial. However, the results were not statistically significant for precision ($p =$). The Wilcoxon test also concluded there was no significant GRU-AE, and GRU-R and GRU-AE models, in terms

Looking only at test batches where there were excel in determining that there are no alerts to be recall. This indicates that the GRU is capable of modeling very well, as alerts are not raised when they should be greater than zero, the ARIMA model outperforms precision. However, this came at a substantial sacrifice likely underpowered. Both of the GRU-based model but only the GRU-AE was able to retain its high recall are reported in Table 3.

表2展示了三种方法的批次性能；每一行展示了能。GRU-AE在所有性能指标（精确率、召回率和M方法最显著的区别在于精确率和召回率上的性能大幅多。在Luminol和GRU-R模型中，与召回率相比，更为保守。然而，在GRU-AE模型中，找到的解决方案 $\mu_P = 0.776$ $\mu_R = 0.938$ ，但并未严重牺牲模型的总够识别导致猪出现呼吸系统疾病症状的环境负面变化变化都会导致呼吸系统疾病的发生。这一点得到了其仅是本次分析中使用的环境指标的证实。

表2的最后一行包含了所有方法在各个批次中的MCC (MCC) 错误分数0 (I)、正确分数1 (C) 或Luminol明显是所有方法中表现最差的，而其他方法低，是ARIMA和GRU-R得分的两倍，几乎是Lumi

从数值上看，我们至少可以看到Luminol和GR Wilcoxon检验的结果显示，在精确率 ($p = 0.184$ 方面，差异没有统计学意义。Wilcoxon检验还得出GRU-AE模型在性能指标方面没有显著差异。

仅观察无事件（表2）的测试批次，深度学习模型的精确率和召回率。这表明GRU能够很好地建模 “模型确实不会发出警报。对于事件数量大于零的批次的模型。然而，这是以召回率大幅牺牲为代价的，表准确率方面性能均有所下降，但只有GRU-AE能够保持

ance of each of the three methodologies; each row d systems when a given batch is used for testing. her methods for all performance metrics measured iffERENCE between the proposed GRU-AE and other nce in terms of precision and recall; ARIMA has also

Luminol and GRU-R models, a higher precision is t these models are more conservative in their alerts. und tend to rely on maximising recall at the expense acRIFICING it so much to be degrading to the model's recall for the GRU-AE model indicates the models ironment that lead to an increased number of pigs r precision of the model could indicate that not all urrences of respiratory disease. This is backed up by ibute to respiratory disease prevalence, not just the

el quantitative summary of the performance of all mber of batches in which each method obtains either f 1 for MCC (C), or in between 0 and 1 (B). In terms rst of the methods, while the others have a similar out as having the lowest count of I, twice as low as times lower than Luminol's, whilst simultaneously

rence in performance between Luminol and GRU-AE lts from the Wilcoxon test showed that the difference = 0.184), recall ($p = 0.059$), and MCC ($p = 0.054$). gnificant difference between the ARIMA model and rms of any of the performance metrics.

ere no events (Table 2), the deep learning models be raised, achieving extremely high precision and odelling the underlying representation of “normality” uld not be. For batches where the number of events rformed both of the GRU-based models in terms of rifice in terms of recall, indicating that the model was dels decreased in performance in terms of precision, recall. The optimised α and β values for each batch

了当使用特定批次进行测试时，三个比较系统的性 MCC)上都优于其他方法。提出的GRU-AE与其他幅提升；ARIMA也实现了这一点，但程度要低得，实现了更高的精确率，这表明这些模型在警报方面方案倾向于以牺牲精确率为代价来最大化召回率，总体性能。GRU-AE模型显著更高的召回率表明其能化；模型的精确率较低可能表明并非所有环境负面其他因素 [42] 也会导致呼吸系统疾病流行，而不仅

的性能高级定量总结。我们统计了每个方法获得或介于0和1之间 (B) 的批次数。在C计数方面，法得分相似 (10–11)。同时，GRU-AE的I计数最inol的四倍，同时B计数也最低。

RU-AE之间的性能差异可能很大。然而，4)、召回率 ($p = 0.059$) 和MCC ($p = 0.054$) 出结论，ARIMA模型和GRU-AE模型、GRU-R和

模型在判断无需发出警报方面表现出色，实现了极高“正常”状态的基础表示，因为当不应发出警报时，次，ARIMA模型在精确率方面优于两种基于GRU表明模型可能能力不足。两种基于GRU的模型在精持其高召回率。每个批次的优化 α 和 β 值见表3。

Table 2. Table of results for the grid search-optimise integrated moving average (ARIMA) model, particle regression, and PSO-optimised threshold-based and many times the respiratory disease prevalence increase test data O_t . The final I/B/C row indicates the num

ID	Events	Lumi	
		P	R
5	0	0.111	0.12
9	0	1.0	0.1
10	0	0.0	0.0
14	0	1.0	1.0
16	0	1.0	1.0
17	0	1.0	0.66
18	0	1.0	1.0
19	0	1.0	0.66
24	0	1.0	1.0
26	4	1.0	0.66
27	5	0.0	0.0
28	3	0.0	0.0
29	1	0.0	0.0
30	0	0.0	0.0
31	1	0.0	0.0
32	0	0.0	0.0
Avg. Events = 0		0.647	0.50
Avg. Events > 0		0.200	0.13
Avg. All		0.507	0.39
I/B/C			

表2。 表为网格搜索优化的领英异常检测库 (Lumin 检测, 以及基于GRU自动编码器产生的马氏距离的P 回率。这些结果来自测试数据 O_t 。最终的I/B/C行表

ID	事件	Lumin	
		P	R
5	0	0.111	0.12
9	0	1.0	0.1
10	0	0.0	0.0
14	0	1.0	1.0
16	0	1.0	1.0
17	0	1.0	0.66
18	0	1.0	1.0
19	0	1.0	0.66
24	0	1.0	1.0
26	4	1.0	0.66
27	5	0.0	0.0
28	3	0.0	0.0
29	1	0.0	0.0
30	0	0.0	0.0
31	1	0.0	0.0
32	0	0.0	0.0
平均事件 = 0		0.647	0.50
平均事件 > 0		0.200	0.13
平均所有		0.507	0.39
I/B/C			

ised implementation of LinkedIn’s anomaly detection library (Lu
particle swarm optimisation (PSO)-optimised threshold-based anom
anomaly detection of Mahalanobis distance produced by a GRU-b
creases from 0 within the batch, P denotes the precision, and R is t
umber of folds that were Incorrect ($MCC = 0.0$), Between ($0.0 < M$

minol			ARIMA			GRU-R			minol			ARIMA			GRU-R		
R	MCC	P	R	MCC	P	R	MCC	P	R	MCC	P	R	MCC	P	R	MCC	P
.125	0.118	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.125	0.118	1.0	1.0	1.0	1.0	1.0	1.0	1.0
0.1	0.316	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.1	0.316	1.0	1.0	1.0	1.0	1.0	1.0	1.0
0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0	0.0
1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
1.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0
.667	0.816	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.667	0.816	1.0	1.0	1.0	1.0	1.0	1.0	1.0
1.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0
.667	0.816	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.667	0.816	1.0	1.0	1.0	1.0	1.0	1.0	1.0
1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
.667	0.816	0.053	0.5	0.162	1.0	0.0	0.0	0.0	.667	0.816	0.053	0.5	0.162	1.0	0.0	0.0	0.0
0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	0.0	0.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0
0.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
0.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
.505	0.551	0.727	1.000	0.727	0.909	0.909	0.909	.505	0.551	0.727	1.000	0.727	0.909	0.909	0.909	0.909	0.909
.133	0.163	0.811	0.500	0.432	0.400	0.400	0.200	.133	0.163	0.811	0.500	0.432	0.400	0.400	0.400	0.200	0.200
.390	0.430	0.753	0.844	0.635	0.750	0.688	0.688	.390	0.430	0.753	0.844	0.635	0.750	0.688	0.688	0.688	0.688
7/5/4			5/1/10			5/0/11			7/5/4			5/1/10			5/0/11		

Table 3. Table of results for the grid search-optimised library (Luminol), the grid search-optimised ARIMA detection of loss incurred from a GRU-based regression detection of the Mahalanobis distance produced by start and length of the time window (in days), respectively the threshold the loss/Mahalanobis distance need optimised using training data O_e ; these results are for

ID	Events	Luminol		AR
		α	β	α
5	0	12	4	12
9	0	12	4	12
10	0	12	4	12
14	0	12	4	12
16	0	12	4	12
17	0	12	4	12
18	0	12	4	12
19	0	12	4	12
24	0	12	4	12
26	4	12	4	12
27	5	12	4	12
28	3	12	4	12
29	1	12	4	12
30	0	12	4	12
31	1	12	4	12
32	0	12	4	12

5.1. Case Study

The following case is presented for batch 2 methodologies applied to the problem. This batch of anomalous events in a single batch, making method to perform well on. The batch was analysed anomaly detection (Figure 6), an ARIMA-powered (Figure 7, a GRU-R-powered, PSO-optimised anomaly detection (Figure 9). The disease prevalence in both analogue and binary form that was used for anomaly detection and the alerts this was the first principal component of the temporal GRU-R and ARIMA models, the mean square error for the GRU-autoencoder, the binary cross-entropy loss recreation of it was used. The final subplot shows the alerts, along with their alert windows.

The two deep learning-based models show similarities as they both capture similar events within the environment peaks and troughs seen in the loss. The Luminol-based occurrences of respiratory disease prevalence, but false positives that it also produces. The GRU-R and in that no alerts are raised, even though the latter has anomalous points. These points rarely align with actual is largely down to not raising many alerts at all. To balance between these solutions, alerting to almost all with some false positives, though far fewer than the

表3。 表为网格搜索优化的领英异常检测库 (Luminol) 的回归损失PSO优化阈值异常检测以及基于GRU自动 α 和 β 分别表示评估警报的时间窗口的起始和长度 阈值以触发警报。它们使用训练数据 O_e 进行优化;

ID	事件	Luminol		ARIMA
		α	β	α
5	0	12	4	12
9	0	12	4	12
10	0	12	4	12
14	0	12	4	12
16	0	12	4	12
17	0	12	4	12
18	0	12	4	12
19	0	12	4	12
24	0	12	4	12
26	4	12	4	12
27	5	12	4	12
28	3	12	4	12
29	1	12	4	12
30	0	12	4	12
31	1	12	4	12
32	0	12	4	12

5.1. 案例研究

以下案例以第27批次为例, 说明了应用于该问题的单个批次中异常事件数量最多, 是方法表现较好的更网格搜索优化的异常检测 (图6)、ARIMA驱动、神经网络反向传播 (GRU-R) 驱动、粒子群优化算法优化的群优化算法优化的异常检测 (图9) 进行了分析。每 prevalence (流行率) 在模拟和二进制形式; 第二产生的警报。对于Luminol模型, 这是温度、湿度 ARIMA模型, 使用预测值与真实值之间的均方误差型重建之间的二元交叉熵损失。最后一个子图显示了

这两种基于深度学习的模型在底层技术方面显示类似的事件, 这在损失曲线中的共同峰值和谷值中得到呼吸系统疾病高发情况进行警报, 但这些警报被其产生 ARIMA模型则面临相反的问题, 即使后者在初看何警报。这些异常点很少与实际事件相符, ARIMA 警报。然而, GRU自动编码器模型在这两种解决方案发生情况都发出了警报, 尽管存在一些误报, 但数量

sed implementation of LinkedIn’s anomaly detection MA model, PSO-optimised threshold-based anomaly session, and PSO-optimised threshold-based anomaly by a GRU-based autoencoder. α and β denote the espectively, for which an alert is assessed, and τ is eded to exceed in order to raise an alert. They are e from the test data O_t .

ARIMA		GRU-R		GRU-AE	
α	β	α	β	α	β
12	1	1	6	1	1
12	1	1	6	1	2
12	1	1	6	1	2
12	1	1	6	1	4
12	1	1	6	3	1
12	1	1	6	1	5
12	1	1	6	3	5
12	1	1	6	1	5
12	1	1	6	1	6
12	1	2	6	3	6
12	4	4	6	1	6
12	1	5	6	8	6
12	2	1	6	2	6
12	1	1	6	4	6
12	1	1	6	1	6
12	1	1	6	8	5

27 exemplifies the characteristics of each of the tch was chosen as it contained the highest number ng it one of the more challenging batches for a sed by a Luminol-powered, grid search-optimised wered, grid search-optimised anomaly detection maly detection (Figure 8), and a GRU-AE-powered, he first subplot for each figure shows respiratory form; the second plots the univariate time series erts raised for it. In the case of the Luminol model, mperature, humidity, and CO₂ time series. For the ror between predicted and ground truth was used; y loss between each input window and the model’s s the results of the anomaly detection for each of the

signs of consistency in the underlying technology, nvironmental data, demonstrated by the common based model correctly produces alerts to some of the ut these are severely outweighed by the number of nd ARIMA models suffer from the inverse problem r appears at first glance to produce some significant actual events, and the success of the ARIMA model . The GRU-autoencoder model, however, strikes a st all of the occurrences of respiratory disease, albeit the Luminol model.

inol) 、网格搜索优化的ARIMA模型、基于GRU 动编码器马氏距离PSO优化阈值异常检测的结果。 度（以天为单位）, τ 是需要超过的损失/马氏距离 ; 这些结果来自测试数据 O_{t_0} .

A		GRU-R		GRU-AE	
α	β	α	β	α	β
12	1	1	6	1	1
12	1	1	6	1	2
12	1	1	6	1	2
12	1	1	6	1	4
12	1	1	6	3	1
12	1	1	6	1	5
12	1	1	6	3	5
12	1	1	6	1	5
12	1	1	6	1	6
12	1	2	6	3	6
12	4	4	6	1	6
12	1	5	6	8	6
12	2	1	6	2	6
12	1	1	6	4	6
12	1	1	6	1	6
12	1	1	6	8	5

题的每种方法的特点。该批次被选中，因为它包含 更具挑战性的批次之一。该批次通过Luminol驱动、 网格搜索优化的异常检测（图7）、门控循环单元- 的异常检测（图8）以及GRU自动编码器驱动、粒子 每个图的第一个子图显示了呼吸系统疾病的 二个子图绘制了用于异常检测的单变量时间序列及其 和CO₂ 时间序列的第一主成分。对于GRU-R和 差；对于GRU自动编码器，使用每个输入窗口与模 了每个警报的异常检测结果及其警报窗口。

示出一致性，因为它们都能在环境数据中捕捉到相 到了体现。基于Luminol的模型能够正确地对部分 产生的大量误报远远超过了。而GRU-R和 看似似乎产生了一些显著的异常点，但并未发出任 A模型的成功很大程度上归功于它几乎不发出任何 案之间取得了平衡，它对几乎所有呼吸系统疾病的 量远少于基于Luminol的模型。

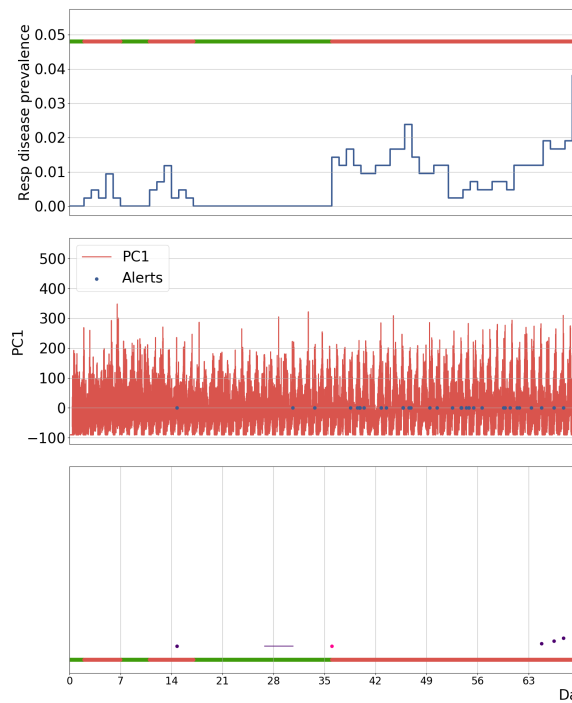


Figure 6. Results for batch 27 processed by the Lu search-optimised parameters α (the start of the win the window), in relation to the prevalence of respira

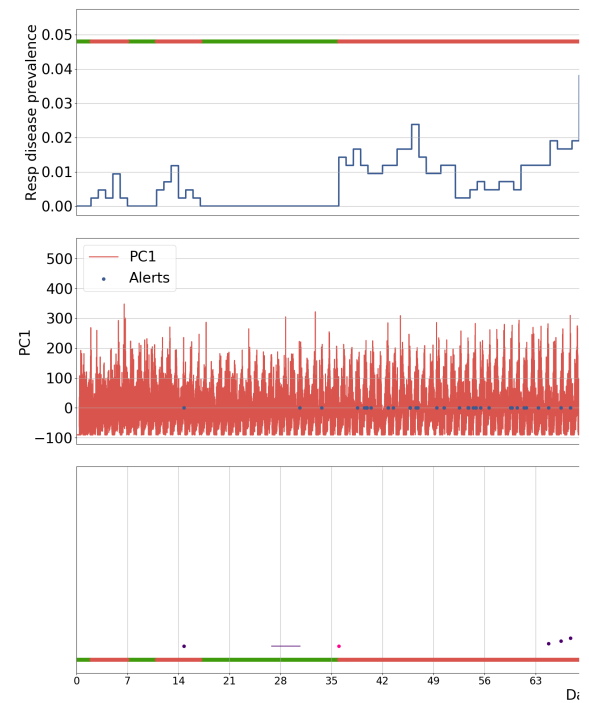


图6. 使用Luminol驱动异常检测处理批次27的结果 (窗口起始位置) 和 β (窗口长度), 与单个批次中呼吸

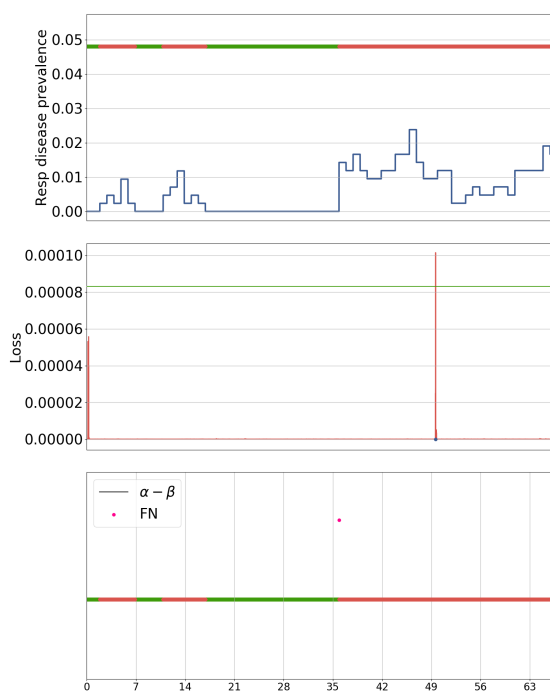


Figure 7. Results for batch 27 processed by the AR search-optimised parameters α (the start of the win window), and τ (the threshold which the loss betwee to be considered an alert), in relation to the prevalen

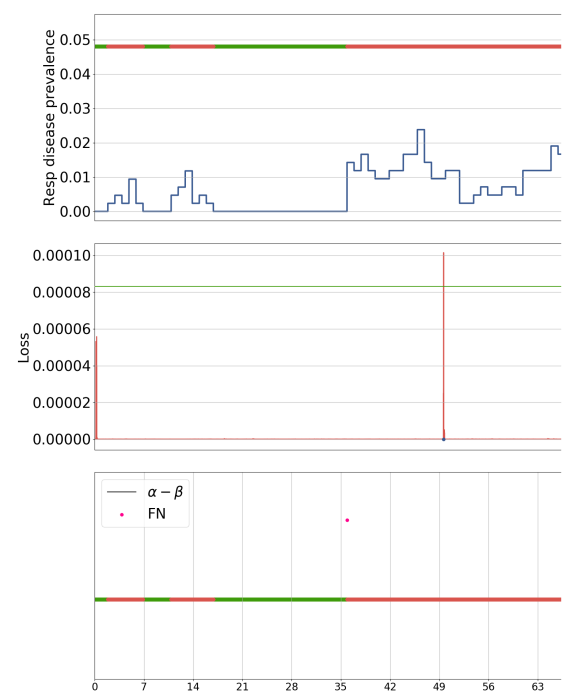
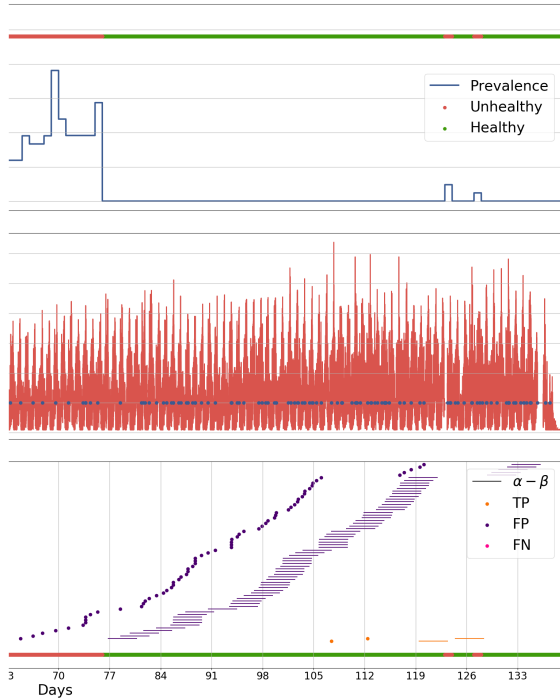
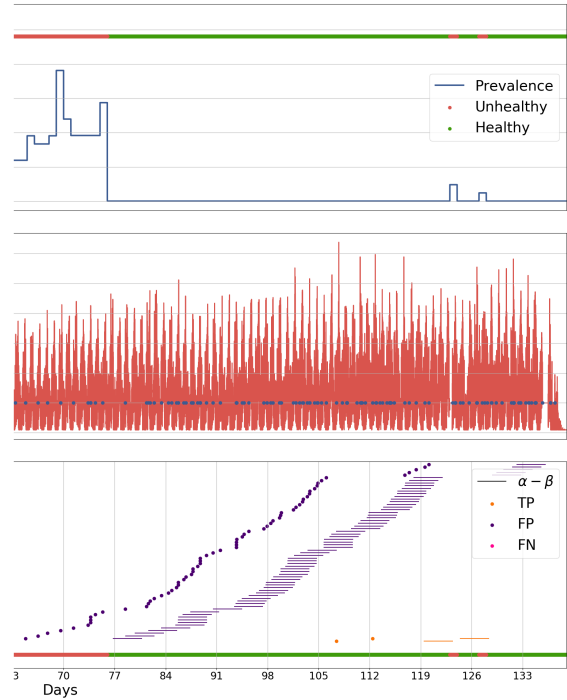


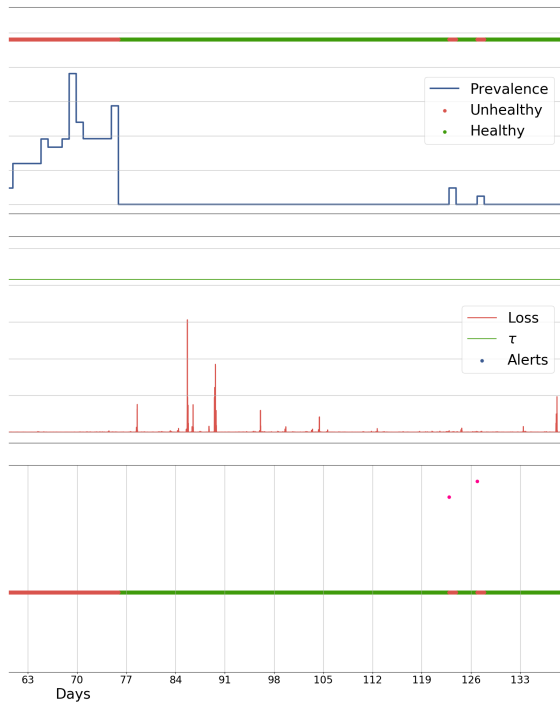
图7. 使用ARIMA驱动异常检测处理批次27的结果 (窗口起始位置)、 β (窗口长度) 和 τ (预测值与实际值单个批次中呼吸系统疾病的流行率之间的关系)。



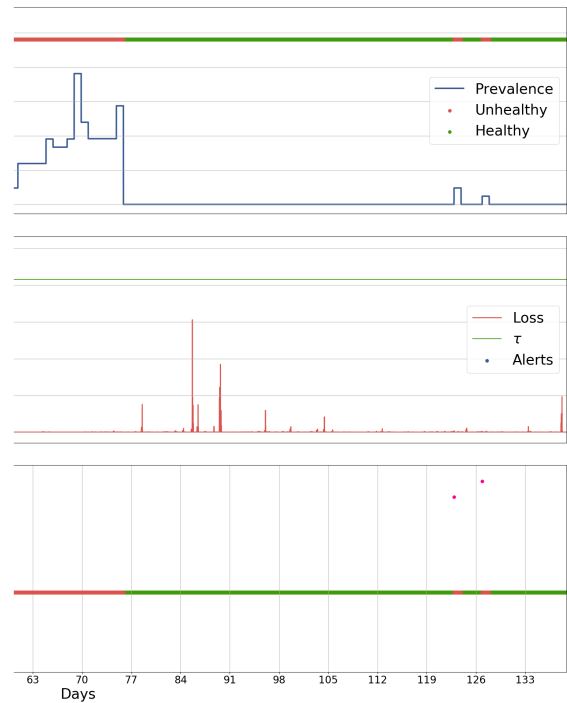
uminol-powered anomaly detection, given its grid indow for which an alert is for) and β (the length of iratory disease within a single batch.



结果，考虑到其通过网格搜索优化的参数 α (警报窗 吸系统疾病的流行率之间的关系。



ARIMA-powered anomaly detection, given its grid indow for which an alert is for), β (the length of the ween predicted values and actual values must cross plence of respiratory disease within a single batch.



果，考虑到其通过网格搜索优化的参数 α (警报窗 值之间的损失必须跨越的阈值，以被视为警报)，与

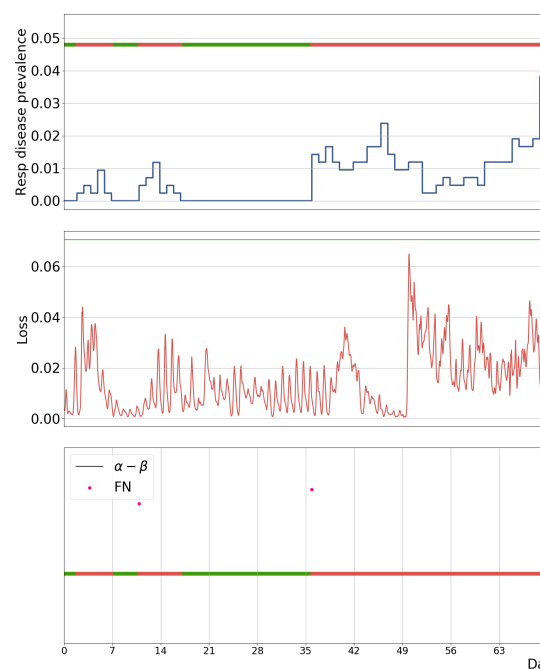
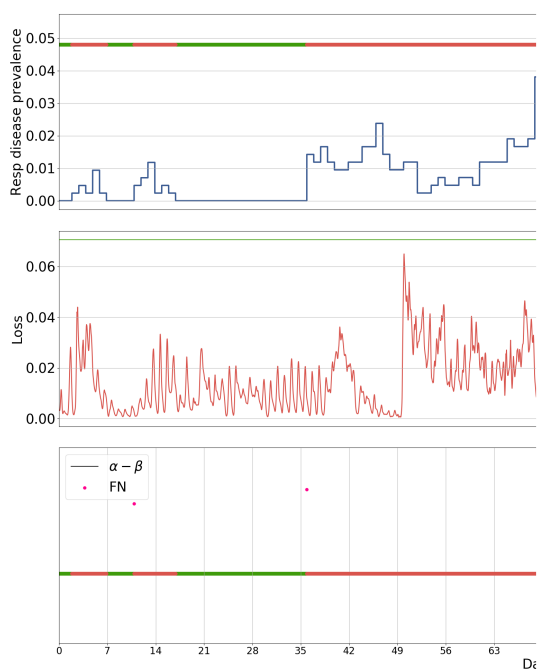


Figure 8. Results for batch 27 processed by the PSO-optimised parameters α (the start of the win the window), and τ (the threshold which the loss b cross to be considered an alert), in relation to the prev

图8. 使用GRU-R驱动异常检测（基于PSO优化参和 τ （预测值与实际值之间的损失必须跨越的阈值以呼吸系统疾病的流行率相关。

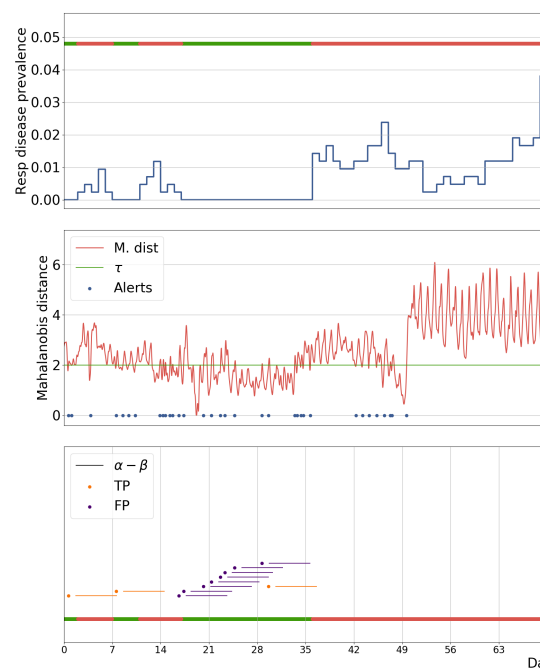
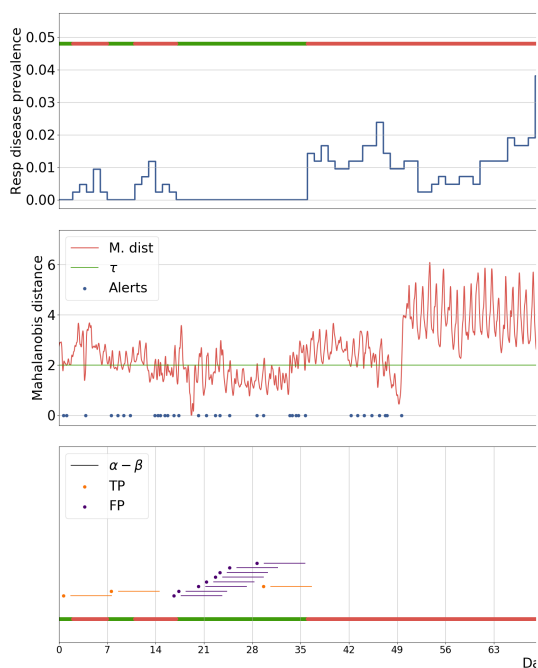
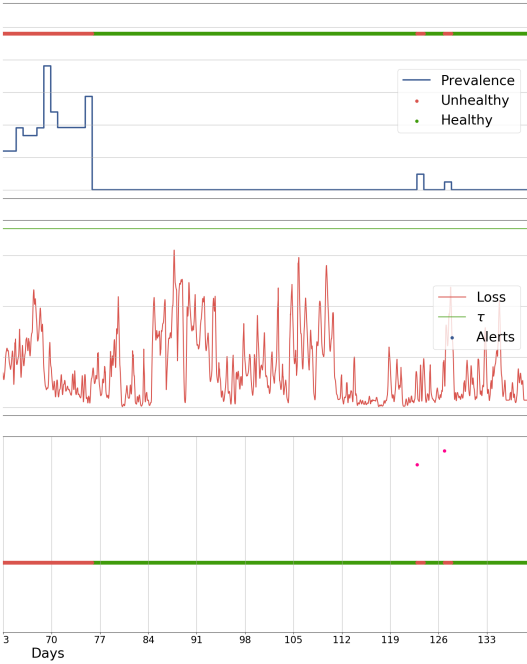
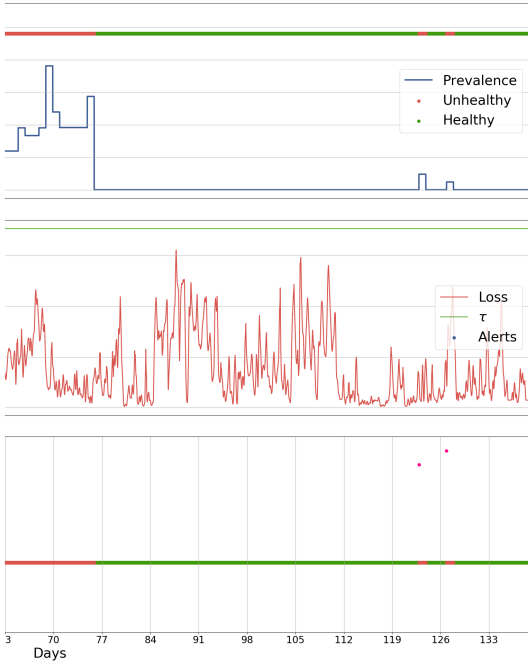


Figure 9. Results for batch 27 processed by the GRU its PSO-optimised parameters α (the start of the win window), and τ (the threshold which the loss betwe to be considered an alert), in relation to the prevalen

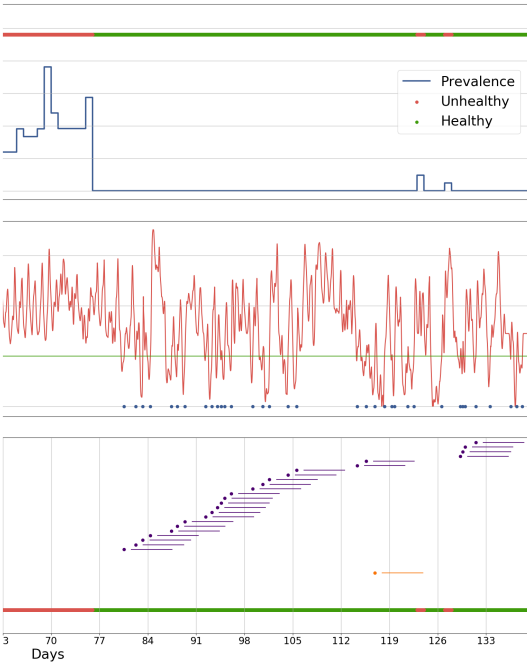
图9. 使用GRU自动编码器驱动异常检测（基于PS口长度）和 τ （预测值与实际值之间的损失必须跨越个批次中呼吸系统疾病的流行率相关。



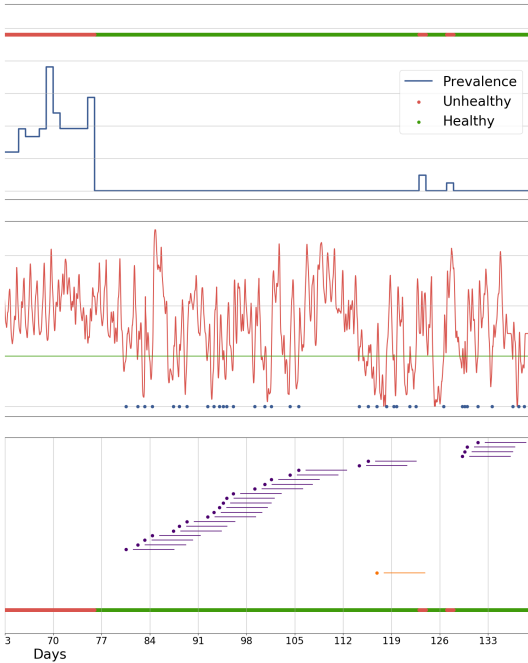
he GRU-R-powered anomaly detection, given its indow for which an alert is for), β (the length of s between predicted values and actual values must relevance of respiratory disease within a single batch.



参数 α (警报窗口的起始位置)、 β (窗口长度) 以被视为警报) 处理批次27的结果, 与单个批次中



RU-autoencoder-powered anomaly detection, given indow for which an alert is for), β (the length of ween predicted values and actual values must cross lence of respiratory disease within a single batch.



PSO优化参数 α (警报窗口的起始位置)、 β (窗 越的阈值以被视为警报) 处理批次27的结果, 与单

5.2. Influence of the Number of Hidden Layers on GRU-AE

The number of layers used in the encoder and decoder is capable of modelling (Figure 10). Since the concept of “healthy” data, the perfect environment’s loss would be the model architecture should be chosen to minimise. Using a small number of layers, 1–15, there was little “healthy data”. Once the number of layers reached 20 and reached its maximum and would not reduce the standard deviation.

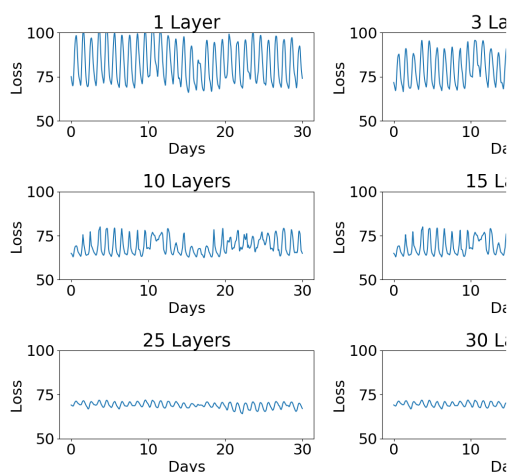


Figure 10. How different numbers of layers in a GRU-AE model affect the modelling of healthy data. The smaller the standard deviation, the better.

5.3. Computation Time

Despite its lacking in performance, the Luminol model was fully trained within a period of 6 h and 24 h. These models were run on a machine containing 44 cores, whereas the two GRU models took a week with an Nvidia GTX 980 Ti.

6. Discussion

This paper set out to develop and evaluate a model comprising GRUs, and to apply the methodology to growing pigs. This network structure was used to the temporal aspect of data within the structure of the data. The GRU-AE specifically could be detected, resulting in a substantially higher recall.

The novelty of this work lies in the complete reassembly of multi-sensor anomaly detection of Malhotra et al. [3] from LSTM cells, currently the most common unit in the network as a whole, as GRUs are less complex and comparable performance [43–46]. A PSO-optimised GRU-AE to determine if a win in methodological changes, this work provides a mu

5.2. 门控循环单元-自动编码器层数对性能的影响

编码器和解码器中使用的层数对GRU-AE能够建模“健康”数据的损失会减少，因此完美环境的损失选择以最小化健康数据中的损失标准差。使用较少的层几乎没有变化。一旦GRU-AE每半部分的层数达到20标准差。

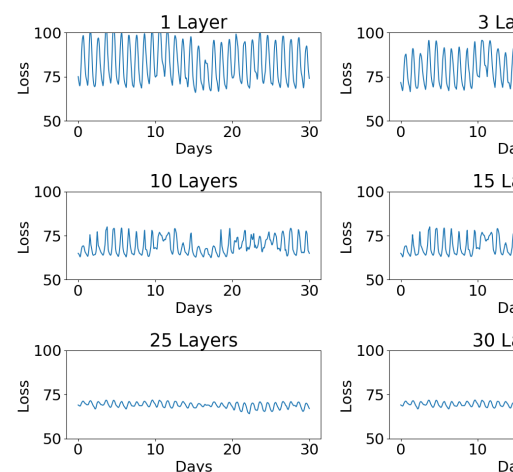


图10. GRU-AE中不同层数如何影响建模环境的能力数据。标准差越小，模型性能越好。

5.3. 计算时间

尽管性能有所欠缺，所提出的基于Luminol的两个GRU模型所需时间的一小部分。基于Luminol MA的模型则需要24小时完成完全训练。这些模型运行在共44个核心的机器上，而两个GRU模型则在一个花费了一周时间。

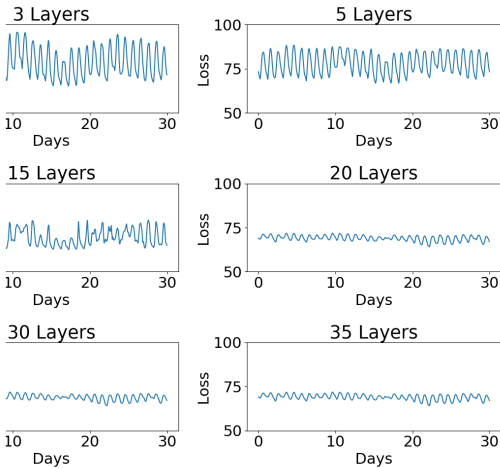
6. 讨论

本文旨在开发并评估一种利用包含门控循环单元猪呼吸系统疾病的早期检测。该网络结构被用于构建中考考虑数据的时序性，而不仅仅是通过数据的结构化特别优于GRU-R，从而为GRU-AE带来了显著更高

这项工作的创新之处在于对 Malhotra 等人进行了全面重新评估。主要调整是将目前循环网络中最都有益，因为 GRU 更简单，因此更高效，同时性能常检测器，用于处理 GRU-AE 的输出，以确定一个还通过案例研究和与其他替代方法的实际比较，对 G 法应用于真实世界数据。

U-AE Performance

nd decoder had an effect on what the GRU-AE was
cept behind the model was that loss is reduced for
ld have a constant loss over time, therefore meaning
ise the standard deviation of the loss in healthy data.
little change in the ability of the GRU-AE to model
ed 20 for each half of the GRU-AE, the performance
standard deviation of loss any further.

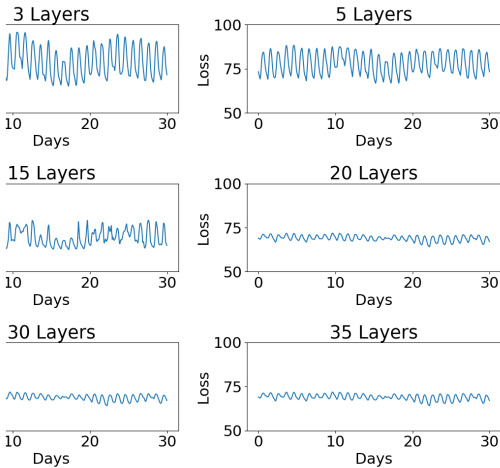


GRU-AE affect the ability to model environmental
ter the model performs.

inol-based model presented (Figure 6) was trained
train the two GRU models presented. The Luminol
d the ARIMA-based model was fully trained within
ntaining two Xeon E5-2699v4 processors (totalling
week on a similarly performing machine equipped

a methodology utilising recurrent neural networks
gy to the early detection of respiratory disease in
to build an autoencoder capable of accounting for
of the network itself, rather than solely through the
outperforms the GRU-R where there were events to
call for the GRU-AE.
eassessment of the idea of a recurrent autoencoder for
. [3]. The primary adjustments made were switching
nit in recurrent networks, to GRUs. This benefits
plex and therefore more efficient whilst providing
ed anomaly detector was also created for processing
indow was anomalous or not. In addition to the
uch more in-depth evaluation of the performance

建模的内容有影响 (图10)。由于模型背后的概念
失在时间上会是恒定的，这意味着模型架构应该选
层数，1-15层，GRU-AE建模“健康数据”的能力
20层，性能达到最大，并且不会进一步降低损失的



力 al

模型 (图6) 的训练和优化所需时间仅为所提出的
ol的模型在6小时内完成完全训练，而基于ARI
运行在一台包含两个Xeon E5-2699v4处理器（总
个性能相似的、配备了Nvidia GTX 980 Ti的机器上

元的循环神经网络的方法，并将该方法应用于生长
建一个自编码器，该自编码器能够在网络结构本身
化来实现。GRU-AE在存在待检测事件的情况下，
的召回率。

出的用于多传感器异常检测的循环自编码器想法进
最常用的 LSTM 单元切换为 GRU。这对整个网络
能相当 [43–46]。此外，还创建了一个 PSO 优化异
个窗口是否异常。除了方法论上的变化，这项工作
GRU-AE 的性能进行了更深入的评估，并将该方

of the GRU-AE through the use of a case study and whilst also applying the methodology to real-world

Given the recent trends in livestock keeping, portions of livestock management and monitoring high-quality, sustainable, and safe meat production [opportunities for automation in disease detection, such to embracing what is generally referred to as precision variety of sensors. Small, relatively cheap, and high variables, such as those assessed in this study, and sensors such as cameras [48] and microphones [49]. I have the potential to lead to epidemics and therefore this reason, a high precision ought to be sacrificed if pose less danger than false negatives in this application upon this, ensuring that recall is maximised as much

The motivation for developing such a method prevalence in the data that was collected. In this work diseases. There were several reasons for this, including diseases was highest amongst this class of pigs in this that affect pigs, such as lameness and digestive and respiratory diseases, developing similar methodology occurrences to validate the model.

Steps were taken to mitigate the challenges posed of respiratory disease (light, mild, severe, and irre justified, as this methodology is for an early-warning disease or if there is not. Therefore, looking at respiratory segmented into levels of severity, makes for a simple being tackled. It should be noted that the methodology respiratory diseases in pigs.

The detection of respiratory disease following a the parameters derived from literature, such that pathogen incubation (e.g., PRRS has an average incubation GRU-AE are able to raise an alert for oncoming respiratory first showing symptoms. The Luminol model does inability to find hyper-parameters that result in a set values for α and β , as there is no repercussion for doing

Due to the renewed popularity of RNN-based exceptional performance in fields such as natural language increase the interpretability of these black-box models and future work might consider applying such method of the risk factors for respiratory disease in growing assumption-free and works without considering details to validate that this multi-sensor anomaly detection scale thus validating its robustness, would be valuable to

7. Conclusions

We have presented a deep learning-based model of respiratory disease in growing pigs. The model combined with a PSO-optimised anomaly detector anomalies that will lead to occurrences of respiratory country-specific environments. This GRU-AE was able

通过案例研究和与其他替代方法的实际比较，对 GR 用于真实世界数据。

鉴于畜牧业近期的趋势，人们正朝着自动化畜牧可持续和安全肉类生产不断增长的需求 [47]。这些畜牧例如这里开发的技术。一种普遍采用的方法，通常被小型、相对便宜且高分辨率的传感器可以监测环境变化传感器捕获大量其他数据 [48] 和麦克风 [49]。在疾病因此会增加畜牧系统的疾病风险。出于这个原因，如也是值得的，因为在这个应用中，假阳性带来的危险点，确保尽可能最大化召回率。

开发这种方法的动机是处理收集到的数据中疾病于呼吸系统疾病的检测。有几个原因促使我们这样做病率最高。我们也有其他影响猪的疾病的数据，例如于呼吸系统疾病，开发类似的方法会导致模型验证的

为应对数据带来的挑战，我们采取了措施，例如和不可恢复）合并，如第3.2节所述。这样做是合理系统只需要知道是否有疾病或没有疾病。因此，将呼吸严重程度分段，可以创建一个更简单、更专注于所解决可以应用于猪的所有类型的呼吸系统疾病。

环境条件变化后呼吸系统疾病的检测符合文献中如，PRRS的平均潜伏期为14天 [32]) 不会预测即将的猪首先出现症状前1-7天发出即将发生呼吸系统疾病这可能是由于其无法找到导致强MCC的超参数。这样做没有后果。

由于RNN-based模型在自然语言处理等领域表法来提高这些黑盒模型的可解释性。该领域的工作已法应用于模型，以基于GRU-AE更好地理解生长猪呼吸考虑国家差异，因此进行额外工作以验证该多传感器棒性，将对更广泛的社区有价值。

7. 结论

我们提出了一种基于深度学习的模型，能够对生长的基于GRU的自动编码器与PSO优化异常检测器相病的异常方面显示出强大的潜力，并且对特定国家环境方法

and concrete comparisons with alternative methods, real data.

g, there is a move towards automating significant in response to the ever-increasing demand for n [47]. These changes in livestock management offer such as the one developed here. A common approach in precision livestock farming (PLF) is the introduction of a high-resolution sensors can monitor environmental and also capture large quantities of other data using []. In the case of disease detection, any false negatives or increase the disease risk of livestock systems. For if a higher recall can be achieved, as false positives reduction. The GRU-AE presented in this paper acts as such as possible.

Methodology was to deal with the very low disease risk, we concentrated on the detection of respiratory including the fact that the prevalence of respiratory in this dataset. We have similar data for other disorders diseases, but as their incidence is much lower than methodologies would result in an insufficient number of

used by the data, such as summing all classifications (recoverable), as described in Section 3.2. This was a simple system, which only needs to know if there is a respiratory disease prevalence as a whole, rather than a simpler dataset that is more focussed on the challenge methodology developed here can be applied to all types of

g a change in the environmental condition fits within at we are not predicting oncoming disease before incubation time of 14 days [32]). Both the GRU-R and respiratory disease 1–7 days prior to pigs in a batch does not reflect this, however, this is likely due to its a strong MCC. This causes the PSO to maximise the doing so if the performance is poor regardless.

used models for sequence analysis, a result of their language processing, there is a need for methods to models. Work in this area has steadily grown [50,51], methods to the model to gain a better understanding in growing pigs based on the GRU-AE. As this method is differences between countries, additional work to scales beyond the domain of health within agriculture, to the wider community.

Model capable of raising an early warning to occurrences posed methodology of a GRU-based autoencoder model shows strong potential in its ability to detect respiratory disease in growing pigs, and it is robust to able to outperform other comparable methodologies

RU-AE 的性能进行了更深入的评估，并将该方法应

畜牧业管理和监测的重大部分发展，以应对对高质量、畜牧业管理的变化为疾病检测自动化提供了机会，被称为精准畜牧业（PLF），是引入各种传感器。变量，例如本研究中评估的变量，并使用摄像头等病检测的情况下，任何假阴性都有可能导致流行病，如果可以实现更高的召回率，即使牺牲精确率，这险小于假阴性。本文提出的GRU-AE正是基于这一

病发病率非常低的问题。在这项工作中，我们专注做，其中包括该数据集中这类猪的呼吸系统疾病发如跛行和消化系统疾病，但由于它们的发病率远低的样本数量不足。

如将所有呼吸系统疾病的分类（轻微、轻度、严重的，因为这种方法是早期预警系统设计的，该呼吸系统疾病发病率作为一个整体来看，而不是按严决问题的数据集。需要注意的是，这里开发的方法

中推导出的参数，即我们在病原体潜伏期之前（例将到来的疾病。GRU-R和GRU-AE均能够在批次中疾病的警报。然而，Luminol模型无法反映这一点，这导致PSO最大化 α 和 β 的值，因为如果性能差，这

表现出色，其用于序列分析的新兴流行性导致需要方已稳步增长 [50,51]，未来的工作可以考虑将此类方呼吸系统疾病的风险因素。由于该方法无假设且不器异常检测是否超出农业健康领域，从而验证其鲁

生长猪呼吸系统疾病的早期发生发出预警。所提出结合的方法，在检测将导致生长猪发生呼吸系统疾病环境具有鲁棒性。这种GRU-AE能够优于其他可比

in precision, recall, and overall performance in both s of disease.

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在精确率、召回率和整体性能方面，无论是在存在疾

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